CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL

JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

BEST FROM KOTA LECTURES <<< CLICK

QUIZES <<< CLICK

REVISION FLASH CARDS &

MANY MORE <<< CLICK

PHYSICS

1) Consider the two idealised systems (i) a parallel plate capacitor with large plates and small separation and (ii) a long solenoid of length L >> R, radius of cross-section. In (i) E is ideally treated as a constant between plates and zero outside. In (ii) magnetic field is constant inside the solenoid and zero outside. These idealised assumptions, however, contradict fundamental laws as below

(1) case (i) contradicts Gauss' law for electrostatic fields

(2) case (ii) contradicts Gauss' law for magnetic fields

(3) case (i) agrees with $\oint E.dl = 0$

(4) case (ii) contradicts $\oint H.dl = I_{en}$

2) Magnetism at the center of a bar magnet is -

(1) Half of the poles

(2) Maximum

(3) Zero

(4) Minimum

3) A loop, made of straight edges has six corners at A (0, 0,0), B (L, 0,0), C (L, L, 0), D (0, L, 0), E (0, L, L) and F (0,0, L).A magnetic field $B = B_0\left(\hat{i} + \hat{k}\right)$ T is present in the region. The flux passing through the loop ABCDEFA (in that order) is

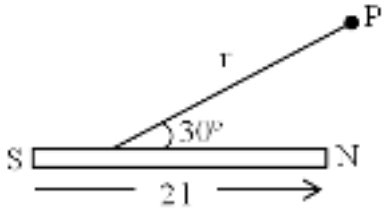
(1) B_0L^2Wb

(2) $2B_0L^2$Wb

(3) $\sqrt{2}B_0L^2$Wb

(4) $4B_0L^2$Wb

4) A short bar magnet of length 2l and magnetic moment 10 A m^2 final the magnetic field at a point 'P' at distance r = 0.1 m from the centre of magnet. The magnet is making an angle 30° with point as shown in figure.



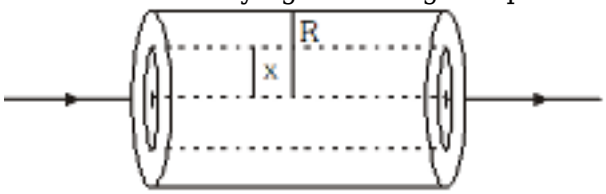
(1) $\sqrt{\frac{7}{4}} \times 10^{-3}T$

(2) $2 \times 10^{-3}T$

(3) $\sqrt{\frac{13}{4}} \times 10^{-3}T$

(4) $10^{-3}T$

5) Current density of an area of cross section is varying according to equation $J = J_0\left(1 - \frac{x}{R}\right)$. Find the current through the cylinder



(1) $\pi R^2 J_0$

(2) $\frac{\pi R^2 J_0}{2}$

(3) $\frac{\pi R^2 J_0}{3}$

(4) $\frac{\pi R^2 J_0}{6}$

6) The core of electromagnets are made of ferromagnetic materials which have

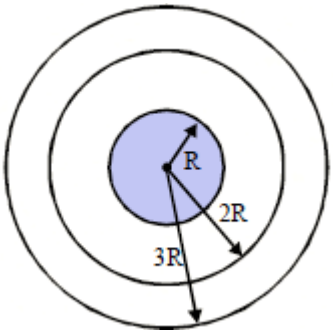
(1) High retentivity and high permeability

(2) High retentivity and low permeability

(3) Low retentivity and high permeability

(4) Low retentivity and low permeability

7) A uniform magnetic field B exists in circular region of radius R as shown which is decreasing with time at a constant rate. There is a concentric wooden ring which have uniformly distributed charge Q and radius 2R and another concentric uncharged copper ring of radius 3R, then select



INCORRECT alternative (Assume rings are placed on frictionless surface)

(1) There is no induced current in wooden ring.

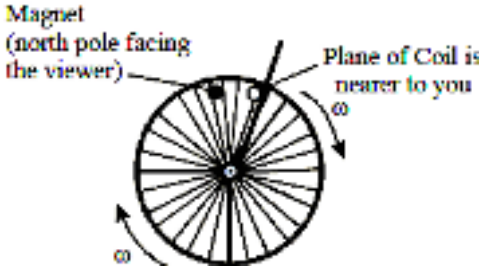
(2) Induced electric field in wooden ring is more in magnitude than copper ring.

(3) Induced electric field in copper ring is more in magnitude than wooden ring.

(4) Wooden ring will start rotating.

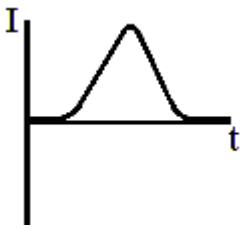
8) In a bicycle speedometer, a bar magnet is attached to the spokes of the wheel and a coil is attached to the frame so that the north pole of the magnet moves past it once for every revolution of the wheel. As the magnet moves past the coil, a pulse of current is induced in the coil. A computer then measures the time between pulses and computes the bicycle's speed. Figure shows the magnet

about to move past the coil. Which of the graphs shows the resulting current pulse ? Take

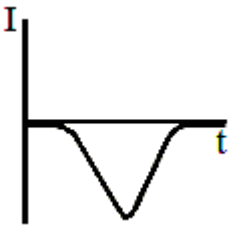


counterclockwise in figure current to be positive.

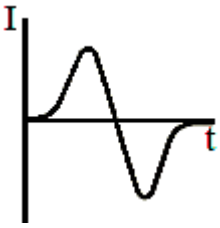
(1)



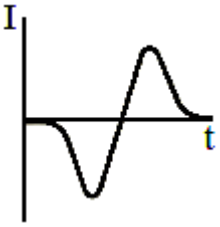
(2)



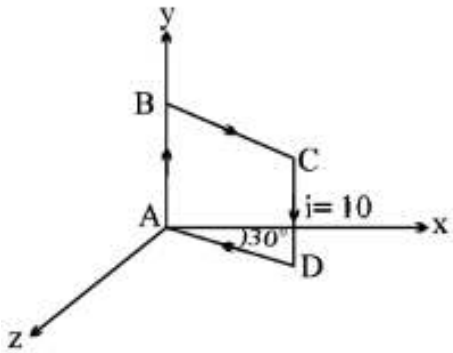
(3)



(4)



9) Figure shows a square current carrying loop ABCD of side 10 cm and current i = 10 A. The



magnetic moment $\vec{M}$ of the loop is

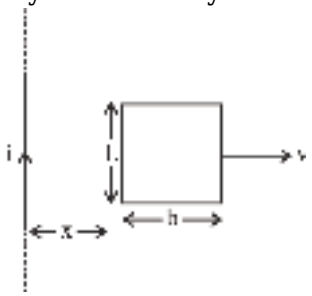
(1) $(0.05)\left(\hat{i}-\sqrt{3}\hat{k}\right)\ A-m^2$

(2) $(0.05)\left(\hat{j}+\hat{k}\right)\ A-m^2$

(3) $(0.05)\left(\sqrt{3}\hat{i}+\hat{k}\right)\ A-m^2$

(4) $\left(\hat{i}+\hat{k}\right)-m^2$

10) A rectangular loop, moves away with velocity = v from a infinite wire of current i. The emf



induced in the loop is given by :-

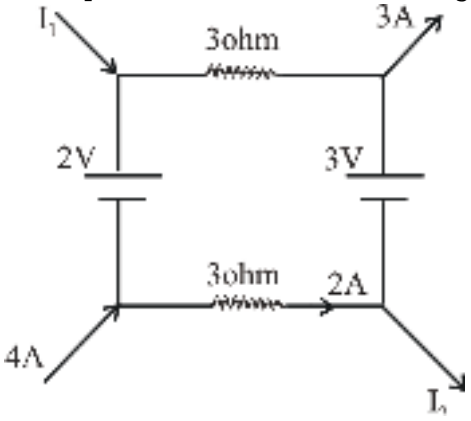
(1) $\frac{\mu_0 I L b v}{2\pi x(x+b)}$

(2) $\frac{2\mu_0 I L b v}{2\pi x(x+b)}$

(3) $\frac{\mu_0 I L b v}{2\pi x b}$

(4) $\frac{2\mu_0 I L b v}{2\pi x b}$

11) A part of circuit is shown in figure. Choose the correct option :-



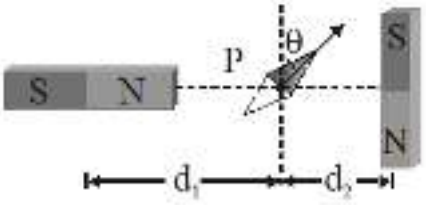
(1) $I_1 / I_2 = 1/2$

(2) $I_1 / I_2 = 2$

(3) $I_1 / I_2 = -1/2$

(4) $I_1 / I_2 = -2$

12) Two magnets A and B are identical and these are arranged as shown in the figure. Their length is negligible in comparison to the separation between them. A magnetic needle is placed between the magnets at point P which gets deflected through an angle θ under the influence of magnets. The



ratio of distance d_1 and d_2 will be :-

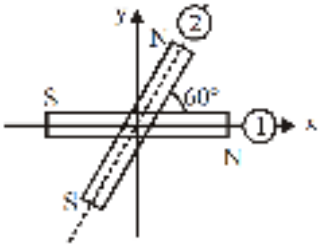
(1) $(2\tan\theta)^{1/3}$

(2) $(2\tan\theta)^{-1/3}$

(3) $(2\cot\theta)^{1/3}$

(4) $(2\cot\theta)^{-1/3}$

13) Two identical bar magnets of magnetic moment 0.1 A–m^2 one located as shown in figure. Magnetic field at point P (0, 0, 1)m is B now if magnet 2 is further rotated by 60° angle in anticlockwise direction then magnetic field at 'P' will be :-



(1) B

(2) $\sqrt{3}B$

(3) $\frac{B}{\sqrt{3}}$

(4) zero

14) Magnetic dipole moment of a bar magnet is 'μ'. If it cuts perpendicular to its length in 1:2 in two parts and both parts combined mutually perpendicular to each other, then net magnetic moment of system is –

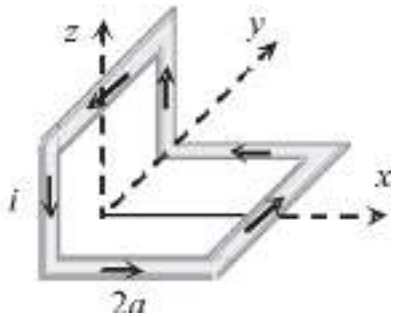
(1) $\sqrt{5}\,\mu$

(2) $\frac{\sqrt{5}\,\mu}{3}$

(3) $\frac{5\,\mu}{3}$

(4) $\frac{5\,\mu}{9}$

15) A non-planar loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length 2a. The magnetic field due to this loop at the point P (a,0,a) points in the direction



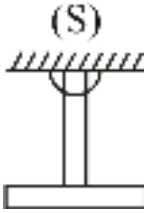
(1) $\frac{1}{\sqrt{2}}\left(-\hat{j}+\hat{k}\right)$

(2) $\frac{1}{\sqrt{3}}\left(-\hat{j}+\hat{k}+\hat{i}\right)$

(3) $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)$

(4) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)$

16) Two identical rods each of mass 'M' and length 'L' are rigidly joined and then suspended into a vertical plane so as to oscillate freely about on axis normal to the plane of paper passing through (S).



Find time period of oscillation.

(1) $\frac{2\pi}{3}\sqrt{\frac{17l}{2g}}$

(2) $2\pi\sqrt{\frac{l}{g}}$

(3) $2\pi\sqrt{\frac{2l}{g}}$

(4) $2\pi\sqrt{\frac{4l}{6g}}$

17) Which of the following statement is correct, if magnet is dropped down an infinitely long vertical copper tube?

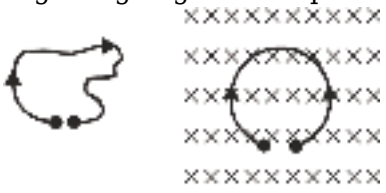
(1) The magnet moves with continuously increasing velocity and acceleration.

(2) The magnet moves with continuously increasing velocity but constant acceleration.

(3) The magnet moves with continuously decreasing velocity and ultimately comes to rest.

(4) The magnet moves with continuously increasing velocity and ultimately acquires a constant terminal velocity.

18) A thin flexible wire of length L is connected to two adjacent fixed points carries a current I in the clockwise direction, as shown in the figure. When system is put in a uniform magnetic field of strength B going into the plane of paper, the wire takes the shape of a circle. The tension in the wire



is:

(1) IBL

(2) $\frac{IBL}{\pi}$

(3) $\frac{IBL}{2\pi}$

(4) $\frac{IBL}{4\pi}$

19) The area of hysteresis curve obtained for a Ferromagnetic material is 0.1 M.K.S. unit. The ferromagnetic substance is of volume $10^{-3}\ m^3$. The heat produced due to energy loss in one cycle of Hysteresis loop, in the substance will be

(1) 10^{-8} J

(2) 10^{-5} J

(3) 10^{-4} J

(4) 10^{-6} J

20) For a moving coil galvanometer, the deflection in the coil is 0.05 rad when a current of 10 mA is passed through it. If the torsional constant of suspension wire is 4.0×10^{-5} Nm rad^{-1}, the magnetic field is 0.01 T and the number of turns in the coil is 200, the area of each turn (in cm^2) is :

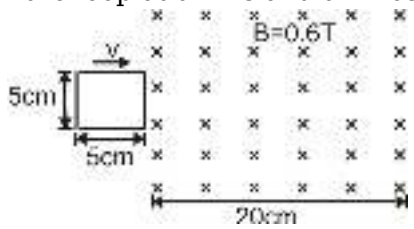
(1) 2.0

(2) 1.0

(3) 1.5

(4) 0.5

21) Figure shows a square loop of side 5 cm being moved towards right at a constant speed of 1cm/sec. The front edge just enters the 20 cm wide magnetic field at t = 0. Find the induced emf in the loop at t = 2s and t = 10s.



(1) 3×10^{-2}, zero

(2) 3×10^{-2}, 3×10^{-4}

(3) 3×10^{-4}, 3×10^{-4}

(4) 3×10^{-4}, zero

22) If two bulbs of wattage 60 W and 100 W respectively each rated at 110 V are connected in series with the supply of 220 V. Which bulb will fuse :-

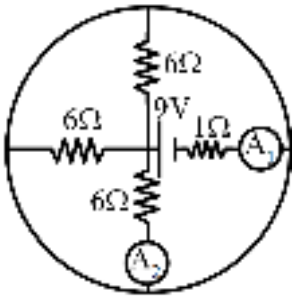
(1) 60 W bulb

(2) 100 W bulb

(3) Both the bulbs

(4) Bulb will not fuse

23) Reading of Ameter A_1 and A_2 will be :-



(1) 3A, 3A

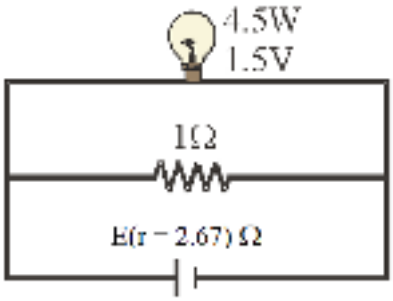
(2) 1A, 3A

(3) 3A, 1A

(4) 1A, 1A

24)

A torch bulb rated as 4.5 W, 1.5 V is connected as shown in the figure. The e.m.f. of the cell needed to make the bulb glow at full intensity is :-



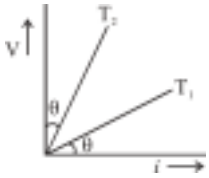
(1) 4.5 V

(2) 1.5 V

(3) 2.67 V

(4) 13.5 V

25) The V-i graph for a conductor at temperatures T_1 and T_2 are as shown in the figure (T_2–T_1) is proportional to :-



(1) cos 2θ

(2) sin θ

(3) cot 2θ

(4) tan θ

26) Temperature of a resistance at temperature t° C is $R = R_0(1 + at + bt^2)$. Here R0 is the temperature at 0°C, the temperature coefficient of resistance at temperature t is:

(1) $\frac{a + 2bt}{1 + at + bt^2}$

(2) (a+2bt)

(3) $\frac{1 + at + bt^2}{a + 2bt}$

(4) constant

27) A Galvanometer of resistance 100 Ω is connected to a battery of 5V along with a resistance of 4900 Ω in series. A full scale deflection of 40 division is obtained in galvanometer. In order to reduce this deflection to 20 division , the additional resistance in series should be :-

(1) 1000 Ω

(2) 9900 Ω

(3) 5000 Ω

(4) 10000 Ω

28) You are provided with 48 cells, each of emf 2 volts and internal resistance 4 ohms. What maximum current can flow in the circuit having an external resistance of 12 Ω ?

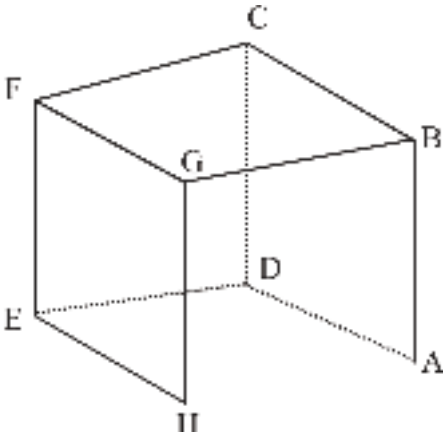
(1) 1 A

(2) 1.2 A

(3) 0.96 A

(4) 1.08 A

29) In the network, if the resistance of each 11 straight wires is R. Find the equivalent resistance between A and H ?



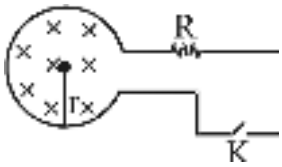
(1) $\frac{2R}{7}$ 0

(2) $\frac{8R}{7}$

(3) $\frac{7R}{5}$

(4) $\frac{2R}{3}$

30) Shown in the figure is a circular loop of radius r and resistance R. A variable magnetic field of induction $B = B_0e^{-t}$ is established inside the coil. If the key (K) is closed, the electrical power

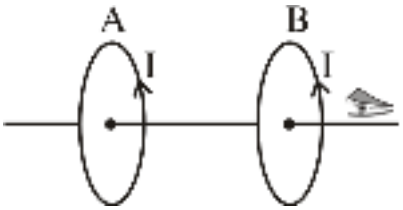
developed right after closing the switch is equal to :- 

(1) $\frac{B_0^2\pi r^2}{R}$

(2) $\frac{B_0 10r^3}{R}$

(3) $\frac{B_0^2\pi^2 r^4 R}{5}$

(4) $\frac{B_0^2\pi^2 r^4}{R}$

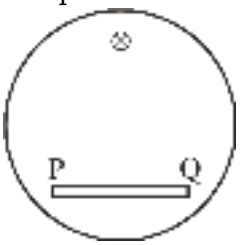
31) Two identical circular coil are carrying current in same direction. If both coils are moved towards each other than current in both coil :- 

(1) Increase

(2) Decrease

(3) Remain same

(4) Increase in A & decrease in B

32) In a cylindrical region uniform magnetic field which is perpendicular to the plane of the figure is increasing with time and a conducting rod PQ is placed in the region. Then 

(1) P will be at higher potential than Q

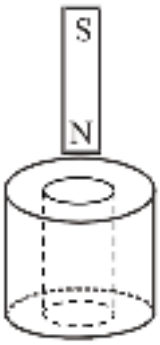
(2) Q will be at higher potential than P

(3) Both P and Q will be equipotential

(4) no emf will be developed across rod as it is not crossing/cutting any line of force

33) When a magnet is released from rest along the axis of a hollow conducting cylinder situated

vertically as shown in figure, then :-



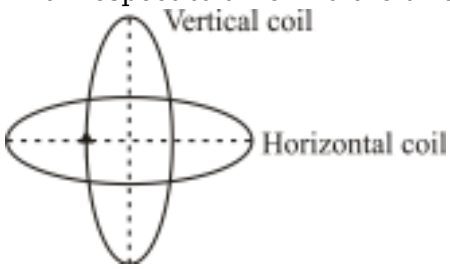
(1) the direction of induced current in the cylinder is clockwise as seen from the above.

(2) the magnet moves with an acceleration more than g = 9.8 m/s^2.

(3) the cylinder gets heated.

(4) the speed of the magnet goes on increasing continuously, if the cylinder is very long.

34) If current in vertical coil increases with respect to time find the direction of induced current in Horizontal coil as seen above the coil.



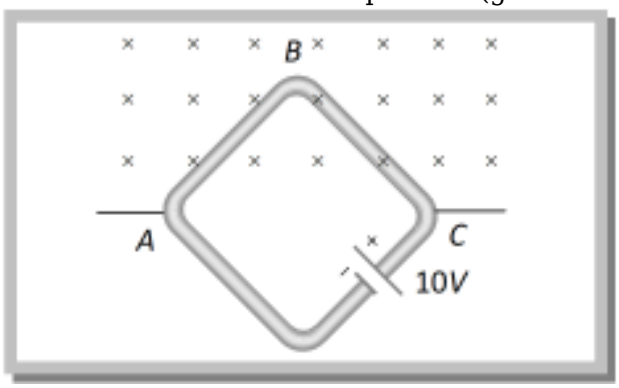
(1) Zero

(2) Clockwise

(3) Anticlockwise

(4) None of the above

35) A fixed square loop of side 1m is placed in a perpendicular magnetic field. Half of the area of the loop inside the magnetic field. A battery of emf 10 V and negligible internal resistance is connected in the loop. The magnetic field changes with time according to relation B = 0.01 - 2t Tesla. The resultant current in the loop will be(given resistance of loop is 1 ohms)



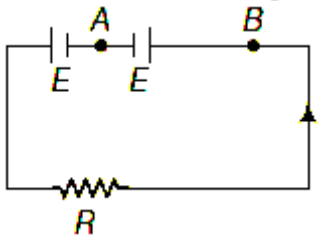
(1) 1 A

(2) 11 A

(3) 10 A

(4) 9 A

36) Two cells of same emf E but internal resistance r_1 and r_2 are connected in series to an external resistor R (figure). What should be the value of R so that the potential difference across the terminals of the first cell becomes zero?



(1) $R = r_1 + r_2$

(2) $R = r_1 - r_2$

(3) $R = 2r_1 + r_2$

(4) $R = \frac{r_1 r_2}{r_1 + r_2}$

37) A toroid of n turns, mean radius R and cross- sectional radius a carries current I. It is placed on a horizontal table taken as xy-plane. Its magnetic moment m

(1) is non zero and points in the z - direction by symmetry

(2) points along the axis of the toroid ($m = m\phi$)

(3) is zero, other wise there would be a field falling as $\frac{1}{r^3}$ at large distances outside the toroid

(4) is pointing radially outwards

38) In a permanent magnet at room temperature

(1) magnetic moment of each molecule is zero

(2) the individual molecules have non-zero magnetic moment which are all perfectly alinged

(3) domains are partially aligned

(4) domains are all perfectly aligned

39)

A paramagnetic sample shows a net magnetisation of 8 Am^{-1} when placed in an external magnetic field of 0.6 T at a temperature of 4 K. When the same sample is placed in an external magnetic field of 0.2 T at a temperature of 16 K, the magnetisation will be

(1) $\frac{32}{3}$ Am^{-1}

(2) $\frac{2}{3}$ Am^{-1}

(3) 6 Am^{-1}

(4) 2.4 Am^{-1}

40) A long solenoid has 1000 turns per metre and carries a current of 1 A. It has a soft iron core of $\propto_r = 1000$. The core is heated beyond the curie temperature, T_c.

(1) The H field in the solenoid is (nearly) unchanged but the B field decreases drastically

(2) The H and B fields in the solenoid are nearly unchanged

(3) The magnetisation in the core reverses direction

(4) None of these

41) A ferromagnetic material is placed in an external magnetic field. The magnetic domains

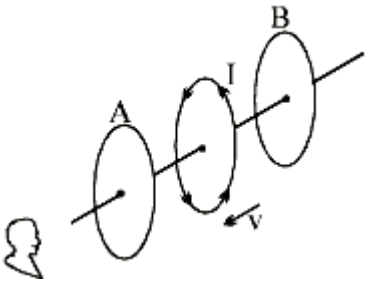
(1) Increase in size

(2) decrease in size

(3) may increase or decrease in size

(4) have no relation with the field

42) Three wire loops and an observer are positioned as shown in the figure. From the observer's point of view, a current I flows counterclockwise in the middle loop, which is moving towards the observer with a velocity v. Loops A and B are stationary. This same observer would notice that



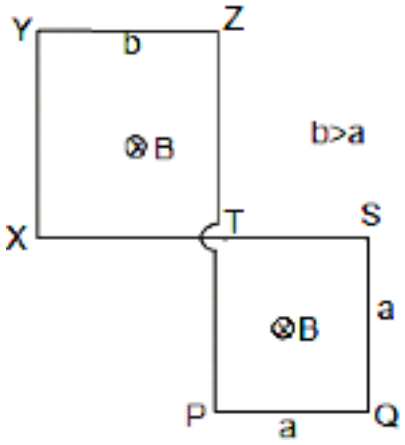
(1) clockwise currents are induced in loops A and B

(2) counterclockwise currents are induced in loops A and B

(3) a clockwise current is induced in loop A, but a counterclockwise current is induced in loop B

(4) a counterclockwise current is induced in loop A, but a clockwise current is induced in loop B

43) Figure shows a wire frame PQSTXYZ placed in a time varying magnetic field given as B = βt, where β is a positive constant. Resistance per unit length of the wire is λ. Find the current induced in the wire and draw its electrical equivalent diagram.



(1) $i = \frac{B}{4\lambda}(b + a)$

(2) $i = 4\lambda B(b + a)$

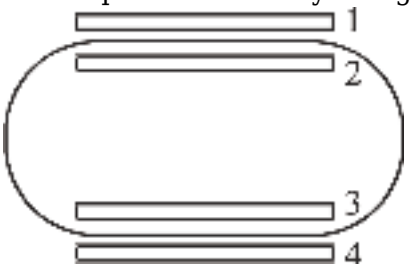
(3) $i = \frac{B}{4\lambda}(2a + b)$

(4) $i = \frac{B}{4\lambda}(b-a)$

44) A magnetising field of 1500 A/m produces a magnetic flux density 0.48 Wb/m^2 in an iron bar. Calculate magnetic susceptibility of iron bar used.

(1) 254
(2) 255
(3) 155
(4) 154

45) Figure shows the path of an electron in a region of uniform magnetic field. The path consists of two straight section, each between a pair of uniformly charged plates and two half-circle. The plates are named 1, 2, 3, & 4 then :-



(1) 1 and 3 at higher (positive) potential and 2 and 4 at lower (negative) potential
(2) 1 and 3 at lower potential and 2 and 4 at higher potential.
(3) 1 and 4 at higher potential and 2 and 3 at lower potential.
(4) 1 and 4 at lower potential and 2 and 3 at higher potential.

CHEMISTRY

1) A, B and C are possible complexes of $CoCl_3.6H_2O$. A shows maximum electrical conductance and C has maximum percentage of nonionisable Cl^- and shows conductivity Then incorrect statement is :-

(1) No weight loss, when A is treated with conc, H_2SO_4
(2) Order of number of moles of AgCl ppt on adding excess $AgNO_3$ solution is A > B > C
(3) Complex C is $[Co(H_2O)_3Cl_3].3H_2O$
(4) Complex B is $[Co(H_2O)_5Cl]Cl_2.H_2O$

2) Which of the following is diamagnetic :-

(1) $[Cr(NH_3)_6]^{3+}$
(2) $[Co(NH_3)_6]^{3+}$
(3) $[Co(NH_3)_6]Cl_2$
(4) $[Cu(NH_3)_4]^{2+}$

3) Correct relation between pairing energy (P) and splitting energy (Δ_0) in complex ion $[Pt(H_2O)_6]^{+4}$:–

(1) $\Delta_0 < P$

(2) $\Delta_0 > P$

(3) $\Delta_0 = P$

(4) Cannot be predicted

4) Among the following compound, which one is both paramagnetic and coloured ?

(1) $K_2Cr_2O_7$

(2) $(NH_4)_2[TiCl_6]$

(3) $VOSO_4$

(4) $K_3[Cu(CN)_4]$

5) Which of the following will have three stereoisomeric forms :-
(i) $[Cr(NO_3)_3(NH_3)_3]$ (ii) $K_3[Co(C_2O_4)_3]$ (iii) $K_3[CoCl_2(C_2O_4)_2]$ (iv) $[CoBrCl(en)_2]$

(1) iii, iv

(2) i, ii and iv

(3) Only iv

(4) All

6)

Which of the following compounds show optical Isomerism :-

(i) Cis-$[Co(NH_3)_4Cl_2]^+$ (ii) trans-$[Co(en)_2Cl_2]^+$
(iii) Cis-$[Co(en)_2Cl_2]^+$ (iv) $[Co(en)_3]^{3+}$

(1) (i) & (ii)

(2) (ii) & (iii)

(3) (iii) & (iv)

(4) (i), (ii) & (iv)

7) An octahedral complex having formula $Ma_2b_2c_2$, which of the following combination is possible to exhibit optical isomerism.

(1) a = Cl^-, b & c = (en)

(2) a & b & c = EDTA

(3) a & b = $C_2O_4^{2-}$, c = en

(4) a = H_2O, b = NH_3, c = Cl^-

8) Select incorrect statement :-

(1) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments.

(2) When $\Delta_0 < P$, then d^6 electron configuration is $t_{2g}^4\, eg^2$

(3) Complex $[Co(OX)_3]^{-3}$ has zero magnetic moment

(4) If the Δ_0 for a Co(III) complex is 18000 cm^{-1} then Δ_t for Co(III) complex with same ligand will be 8000 cm^{-1}

9) Magnetic moment (spin only) of octahedral complex having CFSE = – 0.8 Δ_0 & surrounded by WFL can be :-

(1) $\sqrt{15}$

(2) $\sqrt{8}$

(3) Both (1) & (2)

(4) NOT

10) In Ziese's salt C = C bond length is :

$$\text{Note : } \left\{\begin{array}{l} C-C \text{ bond length in ethane is } 1.54\text{Å} \\ C=C \text{ bond length in ethane is } 1.34\text{Å} \\ C\equiv C \text{ bond length in ethane is } 1.20\text{Å} \end{array}\right\}$$

(1) 1.37 Å

(2) 1.19 Å

(3) 1.87 Å

(4) 1.34 Å

11) Which of the following complex ion is not expected to absorb visible light :-

(1) $[NiCl_4]^{-2}$

(2) $[Ni(CN)_4]^{-2}$

(3) $[Cr(NH_3)_6]^{+3}$

(4) $[FeF_6]^{-3}$

12) **Assertion**: Actinoids exhibits greater range of oxidation state than Lanthanoids.
Reason: Energies of 5f, 6d and 7s are comparable.

(1) Both **Assertion** and **Reason** are true and **Reason** is correct explanation of **Assertion**

(2) Both **Assertion** and **Reason** are true and **Reason** is not correct explanation of **Assertion**

(3) **Assertion** is correct and **Reason** is incorrect

(4) Both **Assertion** and **Reason** are not correct

13) Which of the following statement is incorrect ?

(1) Oxidising power increases in the order $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$

(2) Cr^{+2} is reducing while Mn^{+3} is oxidising agent

(3) Cu has a unique behaviour having a negative $E\Theta$ (M^{+2}/M)

(4) Zn is not a transition element

14) Select the correct statement :-
(a) Mn^{+3} act as stronger oxidising agent than Cr^{+3} & Fe^{+3}
(b) MnF_7, SiF_6^{-2} & BF_6^{-3} all are known flouride
(c) La^{+3} & Lu^{+3} both are diamagnetic
(d) When $K_2Cr_2O_7$ react with H_2O_2 in acidic medium it forms dark orange (CrO_5) compound

(1) a,b,d

(2) a,b,c

(3) a,c

(4) a,b,c,d

15) $KMnO_4$ can't be acidified by ?

(1) Dil. HCℓ

(2) Con. H_2SO_4

(3) Con. HNO_3

(4) All of these

16) When copper is heated with conc. HNO_3 it produces

(1) $Cu(NO_3)_2$ and NO_2

(2) Cu $(NO_3)_2$ and NO

(3) $Cu(NO_3)_2$, NO and NO_2

(4) $Cu(NO_3)_2$ and N_2O

17) Which of the following does not give chromyl chloride test :-

(1) NaCl

(2) KCl

(3) $HgCl_2$

(4) NH_4Cl

18) Which of the following statement is not true ?

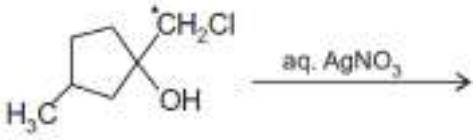
(1) $K_2Cr_2O_7$ solution in acidic medium is orange

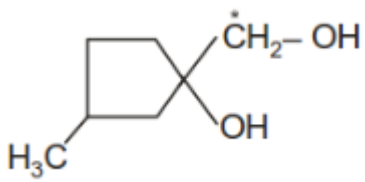
(2) $K_2Cr_2O_7$ solution becomes yellow on increasing the pH beyond 7

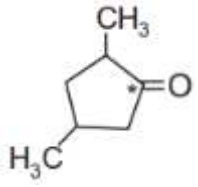
(3) On passing H_2S through acidified $K_2Cr_2O_7$ solution, a milky colour is observed

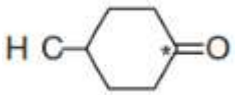
(4) The bond angle of Cr-O-Cr < O-Cr-O in $K_2Cr_2O_7$

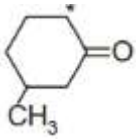
19) The major end product of the following reaction is

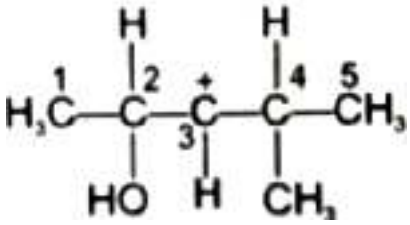


(1) 

(2) 

(3) 

(4) 

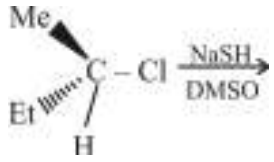
20) In the following carbocation; H/CH_3 that is most likely to migrate to the positively charged carbon is 

(1) CH_3 at C - 4

(2) H at C - 4

(3) CH_3 at C - 2

(4) H at C - 2

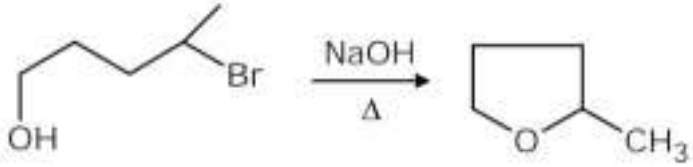
21) For the reaction, 

(1) A single stereoisomer with same configuration will be formed.

(2) A racemic mixture will be produced.

(3) Meso product is formed

(4) A single stereoisomer will be formed.

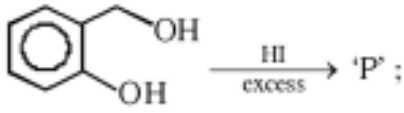
22) The following reaction is described as 

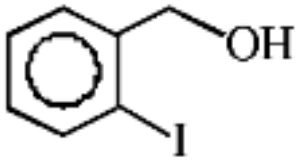
(1) SN_1 reaction with racemisation

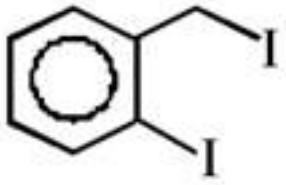
(2) Intramolecular SN_2 reaction with Walden inversion

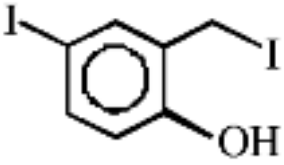
(3) Intramolecular SN_2 reaction with retention of configuration

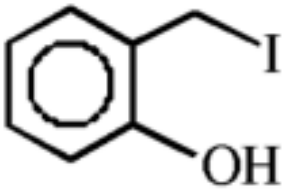
(4) Intramolecular SN_1 reaction with racemisation

23)  Major product 'P' is :-

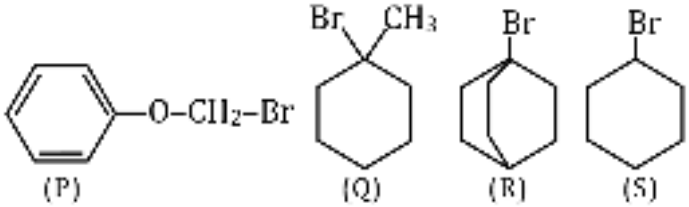
(1) 

(2) 

(3) 

(4) 

24) Rate of reaction of the given compounds with aqueous ethanol follows the order :



(1) P > Q > S > R

(2) Q > P > R > S

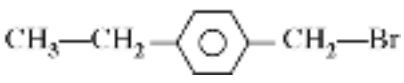
(3) P > R > Q > S

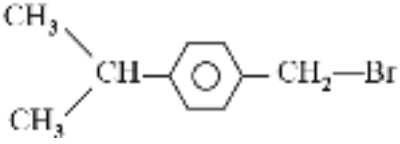
(4) R > P > S > Q

25) Arrange the following compounds in order of decreasing rate of hydrolysis for S_N1 reaction:

(I) 

(II) 

(III) 

(IV) 

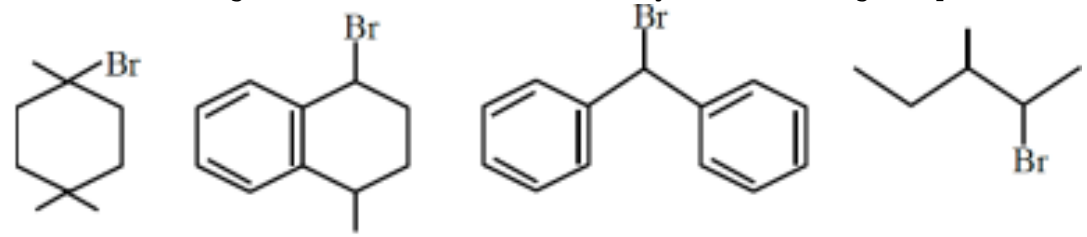
(1) II > III > IV > I

(2) IV > III > II > I

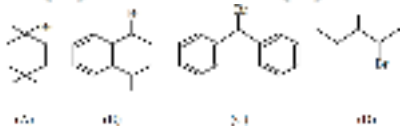
(3) III > IV > II > I

(4) I > II > III > I

26) The ascending order of relative rate of solvolysis of following compounds is



(A) (B) (C) (D)



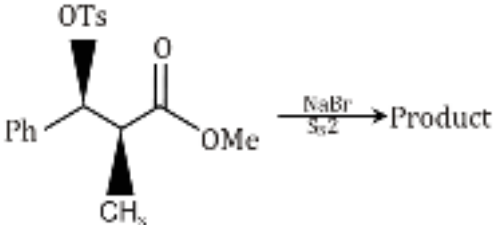
(1) (D) < (A) < (B) < (C)

(2) (C) < (B) < (A) < (D)

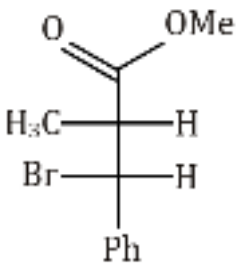
(3) (D) < (B) < (A) < (C)

(4) (C) < (D) < (B) < (A)

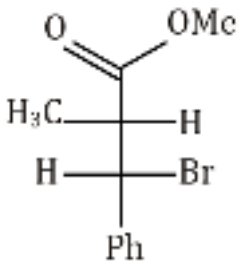
27) What is the product of the following S_N2 reaction?

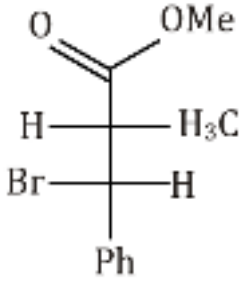


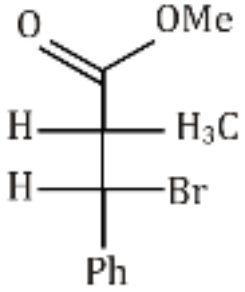
(1)



(2)



(3) 

(4) 

28) For the reaction:

$$RCH_2Br + I^- \xrightarrow[\text{major}]{\text{Acetone}} RCH_2I + Br^-$$

The correct statement is :

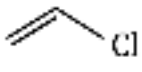
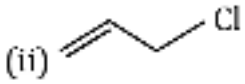
(1) The transition state formed in the above reaction is less polar than the localised anion.

(2) The reaction can occur in acetic acid also.

(3) The solvent used in the reaction solvates the ions formed in rate determining step.

(4) Br– can act as competing nucleophile.

29) Identify correct reactivity order for S_N2 reaction

(i)  (ii)  (iii) 

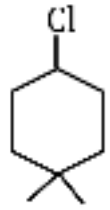
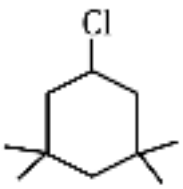
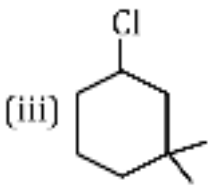
(1) (i) > (ii) > (iii)

(2) (ii) > (iii) > (i)

(3) (i) > (iii) > (ii)

(4) (iii) > (ii) > (i)

30) Identify correct reactivity order for S_N2 reaction

(i)  (ii)  (iii) 

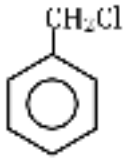
(1) (i) > (ii) > (iii)

(2) (i) > (ii) > (iii)

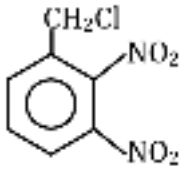
(3) (i) > (iii) > (ii)

(4) (iii) > (i) > (ii)

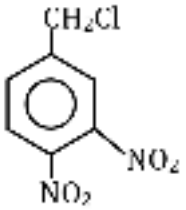
31) The decreasing order of reactivity of the following compounds towards nucleophilic substitution (S_N2) is :



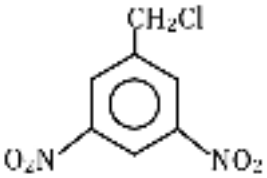
(I)



(II)



(III)



(IV)

(1) (IV) > (II) > (III) > (I)

(2) (II) > (III) > (IV) > (I)

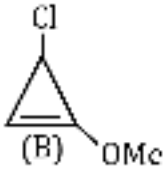
(3) (II) > (III) > (I) > (IV)

(4) (III) > (II) > (IV) > (I)

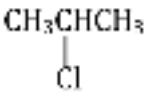
32) The decreasing order of reactivity of the following organic molecules towards $AgNO_3$ solution is :



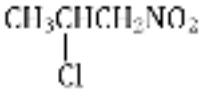
(A)



(B)



(C)



(D)

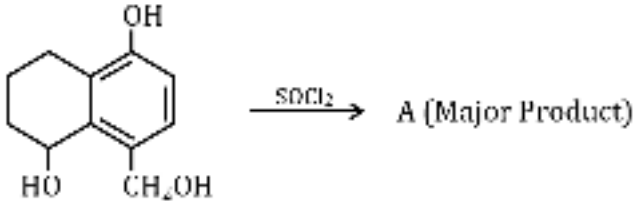
(1) (A) > (B) > (D) > (C)

(2) (A) > (B) > (C) > (D)

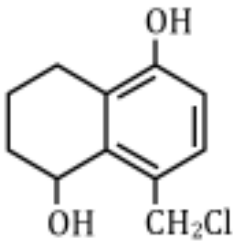
(3) (C) > (D) > (A) > (B)

(4) (B) > (A) > (C) > (D)

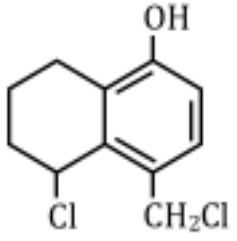
33) Identify A in the given reaction.

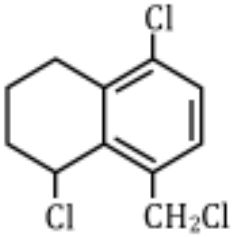


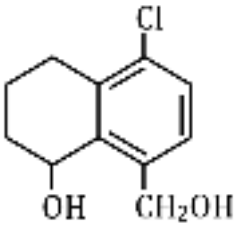
(1)



(2)



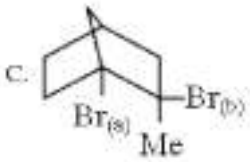
(3) 

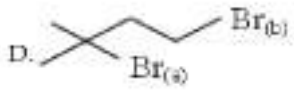
(4) 

34) Choose the halogen which is most reactive towards S_N1 reaction in the given compounds (A, B, C & D).







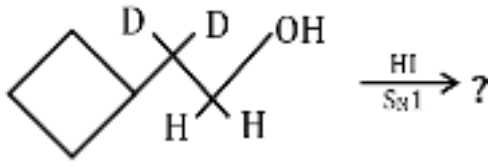


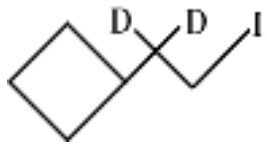
(1) A-$Br_{(b)}$; B-$I_{(b)}$; C-$Br_{(b)}$; D-$Br_{(b)}$

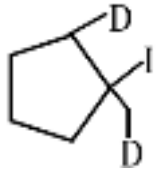
(2) A-$Br_{(a)}$; B-$I_{(a)}$; C-Br(b) ; D-$Br_{(a)}$

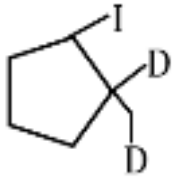
(3) A-$Br_{(a)}$; B-$I_{(a)}$; C-$Br_{(a)}$; D-$Br_{(a)}$

(4) A-$Br_{(b)}$; B-$I_{(a)}$; C-$Br_{(a)}$; D-$Br_{(a)}$

35) 

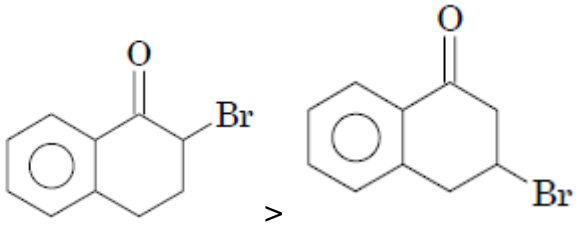
(1) 

(2) 

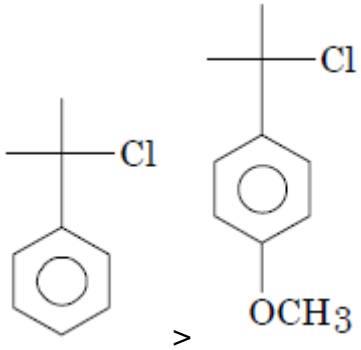
(3) 

(4) None of these

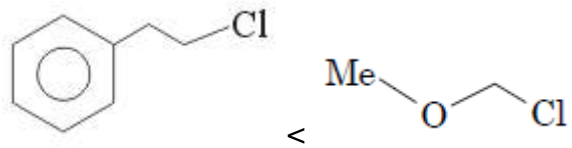
36) Correct order for S_N1 rate :

(1) 

>

(2) 

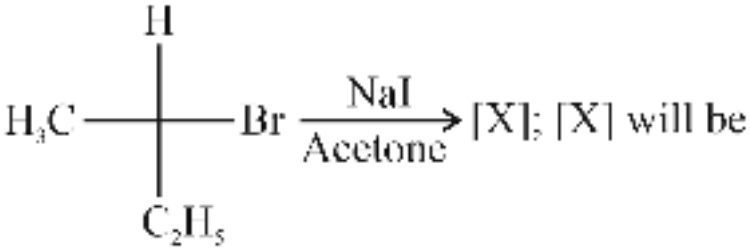
>

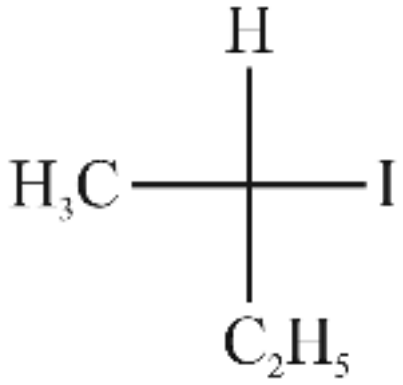
(3) 

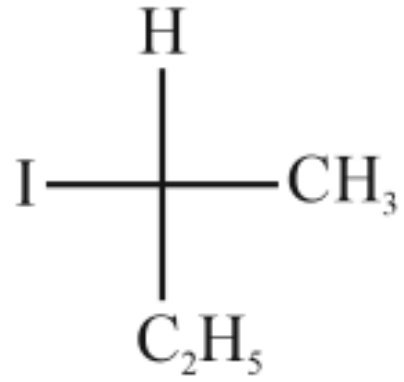
<

(4) All are correct

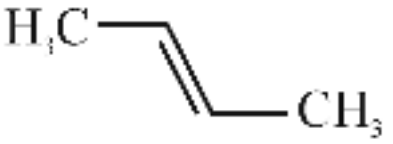
37) In the given reaction,



(1) 

(2) 

(3) Mixture of (A) and (B)

(4) 

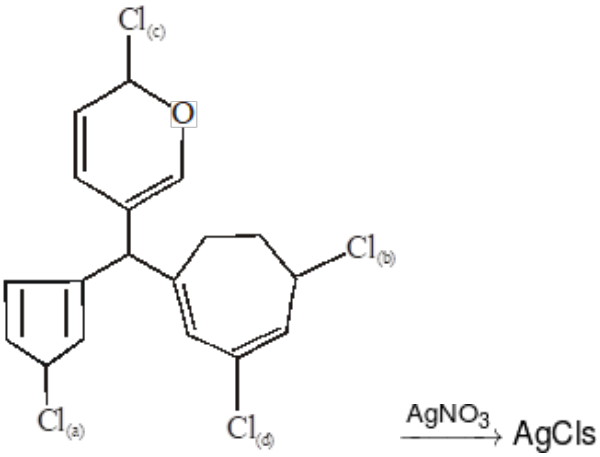
38) Which is correct nucleophilic strength of following nucleophile –

(1) $\overset{\ominus}{O}H > CH_3CO\overset{\ominus}{O} > \overset{\ominus}{O}CH_3 > C_6H_5\overset{\ominus}{O}$

(2) $CH_3\overset{\ominus}{O} > CH_3CO\overset{\ominus}{O} > C_6H_5\overset{\ominus}{O} > \overset{\ominus}{O}H$

(3) $C_6H_5\overset{\ominus}{O} > \overset{\ominus}{O}H > CH_3\overset{\ominus}{O} > CH_3CO\overset{\ominus}{O}$

(4) $CH_3\overset{\ominus}{O} > \overset{\ominus}{O}H > C_6H_5\overset{\ominus}{O} > CH_3CO\overset{\ominus}{O}$

39) 

Which Cl will eliminate with fastest rate in form of $Cl^{\ominus}$ to form AgCl ?

(1) $Cl_{(c)}$

(2) $Cl_{(b)}$

(3) $Cl_{(a)}$

(4) $Cl_{(d)}$

40) The relative reactivities of acyl compounds towards nucleophilic substitution are in the order of -

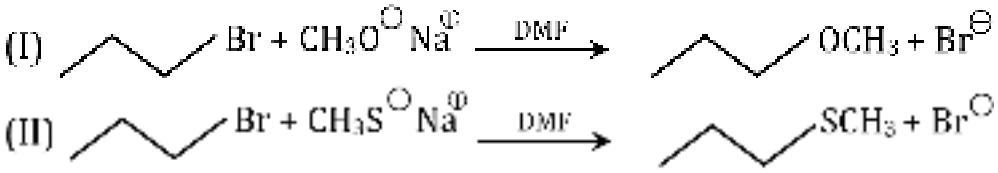
(1) Acid anhydride >Amide > Ester >Acyl chloride

(2) Acyl chloride >Ester>Acid anhydride > Amide

(3) Acyl chloride > Acid anhydride > Ester > Amide

(4) Ester > Acyl chloride > Amide >Acid anhydride

41) Which reaction is faster (I) or (II) and what is its mechanism?



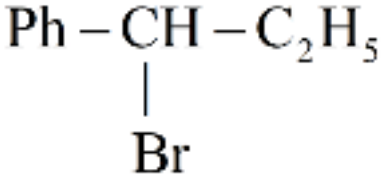
(1) reaction (I), S_N1

(2) reaction (II), S_N2

(3) reaction (II), S_N1

(4) reaction (I), S_N2

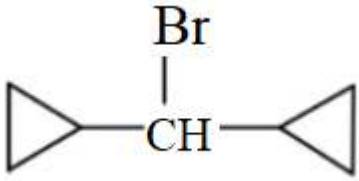
42) Arrange the following compounds in decreasing order of their reactivity for hydrolysis reaction.



(I) $Ph-CH_2-Br$ (II)



(III)



(IV)

(1) I > II > III > IV

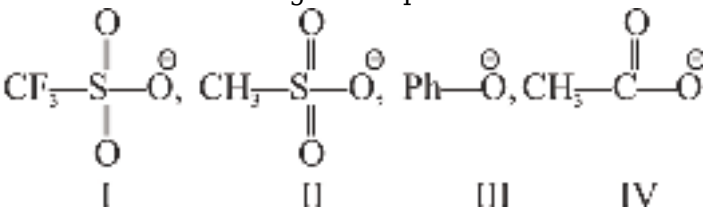
(2) IV > II > I > III

(3) III > IV > II > I

(4) IV > III > II > I

43)

Consider the following nucleophiles



I II III IV

when attached to sp^3-hybridized carbon, their leaving group ability in nucleophilic substitution reactions decreases in the order :-

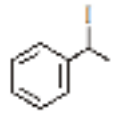
(1) I > II > III > IV

(2) I > II > IV > III

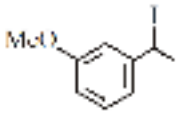
(3) IV > I > II > III

(4) IV > III > II > I

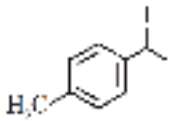
44) Increasing rate of S_N1 reaction in the following compounds is :



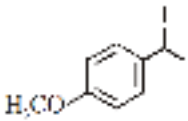
(A)



(B)



(C)



(D)

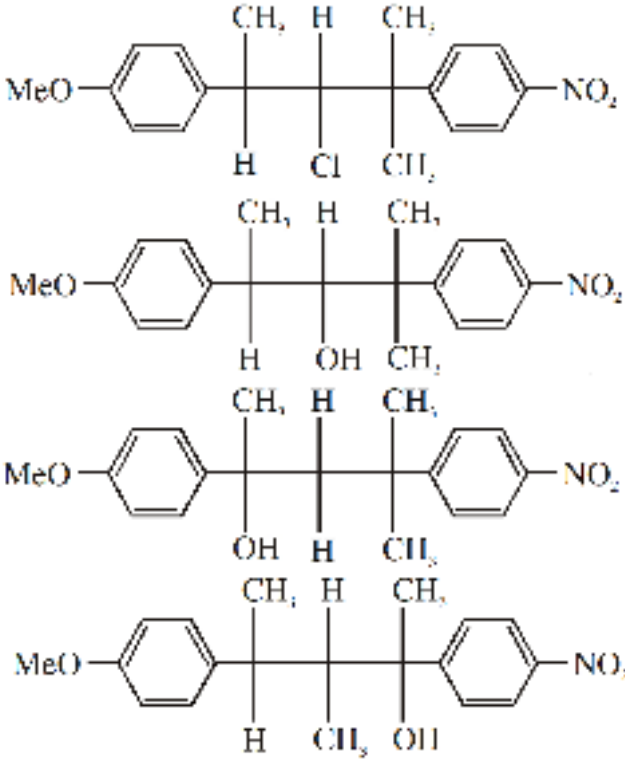
(1) (A) < (B) < (C) < (D)

(2) (B) < (A) < (D) < (C)

(3) (B) < (A) < (C) < (D)

(4) (A) < (B) < (D) < (C)

45) The following compound on hydrolysis in aqueous acetone will give:



It mainly gives

(1) K and L

(2) Only K

(3) L and M

(4) Only M

BIOLOGY

1) Which of the folowing two statements **correct** for kidney transplant ?
(a) Immuno suppresant drugs used for long time inspite kidney properly transplanted
(b) Graft rejection is due to cell mediated immunity

(c) B-cells responsible for graft rejection
(d) Acceptance and rejection of graft depends on specific interferones.

(1) (a) and (b)
(2) (b) and (d)
(3) (c) and (d)
(4) (a) and (d)

2) Incorrect about spleen is -

(1) Secondary lymphoid organ
(2) Bean - shaped
(3) Contains phagocytes only
(4) Traps blood - borne micro organisms

3) Identify the diseases which spread through contaminated food and water?
a. Amoebiasis
b. Pneumonia
c. Common cold
d. Ascariasis
e. Typhoid

(1) a, b, d, e
(2) b, d, e
(3) a, d, e
(4) All the above

4) Which of the following is incorrect matching ?

(1) Typhoid – Sustained high fever
(2) Ascariasis – Blockage of the pulmonary passage
(3) Elephantiasis – Chronic inflammation
(4) Amoebiasis – Abdominal pain and cramps

5) Advantages of using MRI technique of detecting cancer of the internal organs over CT is

(1) Generate 3D image, uses magnetic fields and non-ionising radiation.
(2) Generate 3D image, uses magnetic fields and ionising radiation
(3) Uses antibodies against cancer specific antigens.
(4) Generate 3D image, uses magnetic fields and X-rays and Y-rays

6) **Assertion (A):** Vaccination provides long-term immunity against diseases.
Reason (R): Vaccines contain preformed antibodies that help in fighting pathogens immediately.

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.

(4) A is false, but R is true

7) Identify the correct match from the columns I, II and III

COLUMN–I	COLUMN–II	COLUMN–III
1. Bacteria	A. Plasmodium	i. Infection of lungs
2. Virus	B. Wuchereria	ii. Fever with chills
3. Protozoan	C. Haemophilus	iii. Inflammation of lower limbs
4. Helminth	D. Rhino virus	iv. Infection of upper respiratory tract

(1) 2–c–i, 1–d–iv, 3–4–ii, 4–b–iii

(2) 3–a–ii, 4–b–iii, 2–c–iv, 1–d–i

(3) 4–b–iii, 3–a–ii, 1–c–i, 2–d–iv

(4) 4–b–iii, 3–a–i, 1–c–ii, 2–d–iv

8) Which one of the following group is not of infectious diseases ?

(1) Cancer, Allergy, Rheumatoid arthritis

(2) Dysentery, Plaque, Diphtheria

(3) Polio, Pneumonia, Tetanus

(4) Smallpox, Ringworm, Common Cold

9) Which antibody shows opsonization ?
(a) IgD (b) IgA (c) IgG
(d) IgM (e) IgE

(1) (a), (b), (c)

(2) (c), (d), (e)

(3) (c), (d)

(4) (a), (e)

10) Match the following :-
(A) OPV (i) Rubella
(B) TAB (ii) Tuberculosis
(C) BCG (iii) Typhoid
(D) MMR (iv) Polio

(1) A–i, B–ii, C–iii, D–iv

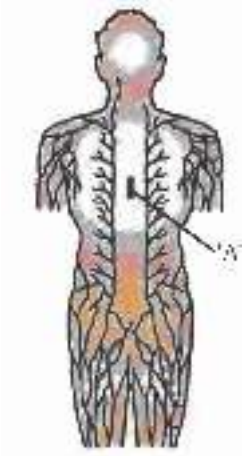
(2) A–ii, B–iii, C–iv, D–i

(3) A–iii, B–iv, C–i, D–ii

(4) A–iv, B–iii, C–ii, D–i

11)

Which of the following statement is false for the structure marked 'A' in the given figure ?



(1) It is related with immunity

(2) It is located near the heart and beneath the breastbone

(3) Here 'B'-lymphocytes differentiates into antigen sensitive lymphocytes

(4) It is included in primary lymphoid organ

12) Common cold differs from pneumonia in that:-

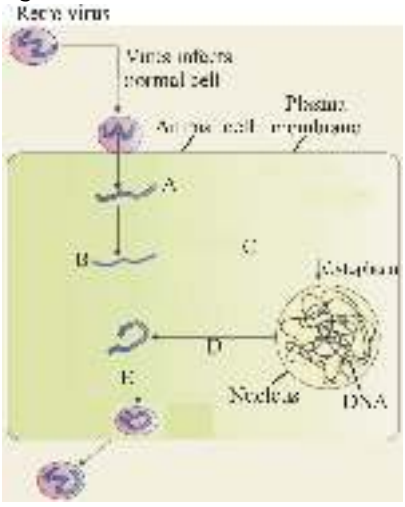
(1) Pneumonia is caused by a virus while common cold is caused by the bacteria.

(2) Pneumonia is infectious disease while common cold is a non-infectious disease

(3) Pneumonia can be treated whereas the common cold has no cure, it can cause death

(4) Pneumonia pathogen infects alveoli whereas the common cold affects nose and respiratory passage but not the lungs.

13) The figure given below shows mode of action of AIDS virus. Identify steps A, B, C, D and E labelled in it.



(1) A-New viral DNA introduced into cell, B-Viral RNA produced, C-Viral DNA incorporated into host genome, D-New viral DNA, E-New viruses produced

(2) A-Viral DNA incorporated into host genome, B-Viral DNA, C-New viral RNA introduced, D-Viral RNA produced, E-New viruses produced

A-Viral RNA introduced, B-Viral DNA,
(3) C-Viral DNA incorporated into host genome, D-New viral RNA produced, E-New viruses produced

(4) A-Viral DNA introduced, B-Viral RNA, C-Viral RNA incorporated into host genome, D-New viral DNA produced, E-New viruses produced

14) Choose the incorrect statements from following:-
(a) Drugs like anti-histamines quickly reduce the symptom of allergy.
(b)In both bone marrow and thymus, immature lymphocytes differentitate into antigen sensitive lymphocyte
(c) Exaggregated response of immune system to certain antigen is called as immunisation.
(d) Lymph nodes, tonsils, payer's patches, appendix are secondary lymphoid organs.

(1) Statement c

(2) Statement c & d

(3) Statement b & d

(4) Statment a & b

15)

The following two organs are examples of secondary lymphoid organs

(1) Spleen and thymus

(2) Lymph nodes and thymus

(3) Spleen and MALT

(4) Bone marrow and thymus

16)

How many statements are incorrect :-
(i) Interferons can prevent neighbour cell from viral infection
(ii) IgM antibody always persist in monomeric form.
(iii) Malaria is a non infectious disease
(iv) Cancer is non communicable.

(1) 4

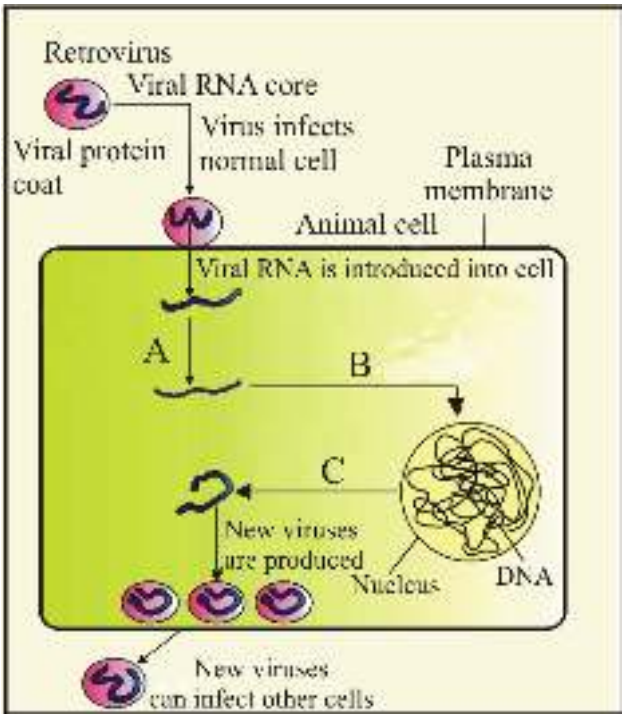
(2) 3

(3) 2

(4) 1

17)

The figure below shows steps A, B & C of replication of retrovirus. Select the option giving correct identification



(1) B–Incorporation of viral DNA into host genome with integrase enzyme

(2) C–Production of new viral DNA by infected cell

(3) A–Replication of viral RNA by protease enzyme

(4) C–Viral RNA replication by reverse transcriptase enzyme

18)

Which of the following are Autoimmune disorders?
A: Myasthenia gravis
B: Rheumatoid arthritis
C: Gout
D: Muscular dystrophy
E: Systemic Lupus Erythematosus (SLE)
Choose the most appropriate answer from the options given below:

(1) A, B, & E only

(2) B, C & E only

(3) C, D & E only

(4) A, B & D only

19) Antigen binding sites on an antibody are formed by :

(1) Variable region of light chain and variable region of heavy chain

(2) Constant region of light chain and variable region of heavy chain

(3) Variable region of light chain and constant region of heavy chain

(4) Constant region of light chain and constant region of heavy chain

20) Which of the following test can be used to identify the causative agent of the symptom listed below?
"Sustained high fever (39^0 to 40^0C),weakness, stomach pain, constipation, headache & loss of

apeptite"

(1) Gravidex test

(2) Mantoux test

(3) Wassermann's test

(4) Widal test

21)

Which of the following is not matched correctly?
(1) α- Interferon ⇒ Biological Response modifier.
(2) MRI ⇒ Use of non-ionising radiation.
(3) Contact inhibition ⇒ Property of cancerous cell.
(4) Carcinogens ⇒ Agents causing transformation of normal cells to cancerous neoplastic cells.

(1) 1

(2) 2

(3) 3

(4) 4

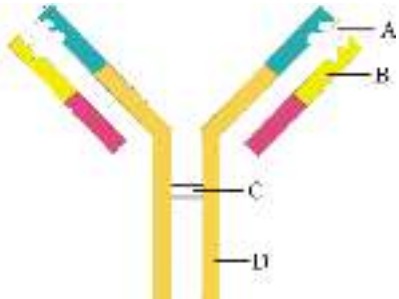
22) Which one of the following are features of acquired immunity. Choose the option with all the correct ones:-
(i) Pathogen specific
(ii) Memory
(iii) Discrimination between self and non-self antigen
(iv) Present at time of birth
(v) Responsible for graft rejection

(1) (i), (ii), (iii), (iv)

(2) (v), (iii), (i), (ii)

(3) (iii), (iv), (v)

(4) (iv), (i), (v)



23) In the given figure, identify the parts labelled as A, B, C & D.

	A	**B**	**C**	**D**
(1)	Epitope	variable region of light chain	disulphide bond	constant region of heavy chain

(2)	Paratope	variable region of light chain	disulplide bond	constant region of heavy chain
(3)	Epitope	constant region of light chain	dipeptide bond	variable region of heavy chain
(4)	Paratope	constant region of light chain	disulphide bond	constant region of heavy chain

(1) 1

(2) 2

(3) 3

(4) 4

24) Which of the following drugs are used in the treatment of allergy and quickly reduce the symptom of allergy ?
(A) Antihistamine (B) Adrenaline (C) Steroid (D) Chloroquine

(1) A only

(2) A and B only

(3) A,B and C only

(4) A,B,C and D

25) Rheumatoid arthritis is an autoimmune disease in which :-

(1) Antibodies are formed against acetylcholine receptor

(2) Anti-histamine is given

(3) Inflammation of synovial membrane occurs

(4) Destruction of muscle occurs

26) Which one of the following sets includes bacterial diseases ?

(1) Tetanus, Tuberculosis, Measles

(2) Diptheria, Leprosy, Plague

(3) Cholera, Typhoid, Mumps

(4) Malaria, Mumps, Poliomyelitis

27) Which of the following disease is auto immune disorder ?

(1) Bronchial Asthma

(2) A.I.D.S

(3) S.C.I.D

(4) I.D.D.M

28) How many of the disease in the list given below are autoimmune disorders :-

Myasthenia gravis, I.D.D.M, osteoarthritis, Albinism, Hemophilia. Hashimoto disease

(1) Six
(2) Three
(3) Four
(4) Five

29) Match the column & choose the correct option.

	Column-I		Column-II
(A)	B-lymphocyte	(i)	Primary and secondary immune response carried out by them
(B)	T-lymphocyte	(ii)	Produce army of protein in response to pathogen
		(iii)	Able to differentiate between self and non self
		(iv)	They do not produce antibodies

	A	B
(1)	i, iii	i, iv
(2)	i, ii, iii	i, iii, iv
(3)	i, ii, iv	ii, iii
(4)	i, iv	ii, iii

(1) 1
(2) 2
(3) 3
(4) 4

30) Which of the following acts as a physiological barrier to the entry of micro organism in human body ?
(A) Epithelium of Respiratory tract
(B) Monocyte
(C) Tear
(D) Skin
(E) Saliva (F) Acid in stomach
(G) N-K-cell

(1) E, F, G
(2) B, C, E
(3) C, E, F
(4) A, B, D

31)

Auto antibodies formed against our self antigen is auto immune disorders. Match the following column-A & B by keep in this in consideration?

	A		B
(i)	Myasthenia Gravis	(a)	Myelin sheath
(ii)	Pernicious Anemia	(b)	β-cells of pancreas
(iii)	IDDM	(c)	Intrinsic castle factor
(iv)	Multiple sclerosis	(d)	Acetyl choline receptor

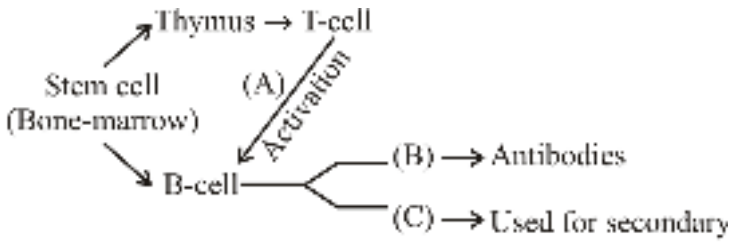
	(i)	(ii)	(iii)	(iv)
(1)	d	c	b	a
(2)	d	c	a	b
(3)	c	d	b	a
(4)	c	b	a	d

(1) 1

(2) 2

(3) 3

(4) 4



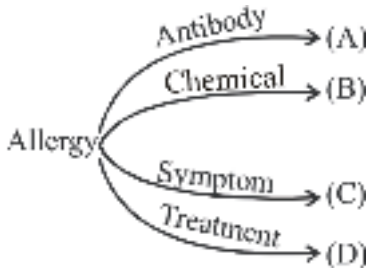
32) Identify A, B and C respectively :-

(1) T-killer cell, plasma cell, T-helper cell

(2) T-helper cell, memory cell, T-plasma cell

(3) T-helper cell, plasma cell, memory cell

(4) T-suppressor cell, memory cell, T-killer cell



33)

	A	B	C	D
(1)	IgA	Heparin	Hypotension	Steroids
(2)	IgE	Heparin	Sneezing	Adrenaline
(3)	IgA	Histamine	Watery eyes	Adreneline
(4)	IgE	Histamine	Running Nose	Steroids

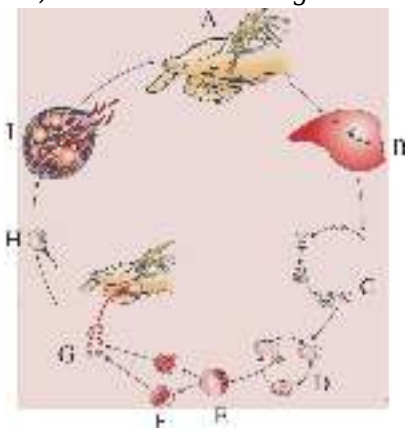
(1) 1

(2) 2

(3) 3

(4) 4

34) Read the following statement and match them with suitable diagram :-



(i) The parasite reproduces asexually in liver cells, bursting the cells and releasing into the blood.
(ii) Sexual stage (gametocytes) develop in red blood cells
(iii) Fertilization and development take place in the mosquito's gut

(1) (i)-D, (ii)-E, (iii)-G

(2) (i)-C, (ii)-F, (iii)-H

(3) (i)-D, (ii)-F, (iii)-H

(4) (i)-C, (ii)-E, (iii)-G

35) How many diseases are caused by bacteria?
Common cold, Pneumonia, Malaria, Dengue, Leshmaniasis, Typhoid, Chikungunya, Tetanus

(1) 4

(2) 3

(3) 6

(4) 2

36) Identify the correct option :-

	Disease	Symptoms	Caused by
(1)	Typhoid	- increase in appetite - Chill and high fever	Salmonella typhi

(2)	Pneumonia and common cold	- cough and headache - in severe cases the lips may turn gray to bluish in color	Streptococcus
(3)	Filariasis	- slowly developing chronic inflammation of the organs	Wuchereria malayi
(4)	AIDS	Progressive decrease in the number of helper T-lymphocytes	Bacterium

(1) 1

(2) 2

(3) 3

(4) 4

37) Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Some human organs like liver, kidney fail to function normally and transplantation is the only remedy to enable the patient to live a normal life.
Reason R : Tissue matching and blood group matching are essential before undertaking any transplant and patient has to take immuno-suppresant thereafter.
In the light of the above statements, **choose** the **correct** answer from the options given below :

(1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

(2) **A** is true but **R** is false.

(3) **A** is false but **R** is true.

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

38) Number of disulphide bonds present on one variable region of light chain and one constant region of heavy chain respectively?

(1) 2,2

(2) 1,4

(3) 4,1

(4) 1, 3

39) How many disease are sexually transmitted disease: Pneumonia, Typhoid, syphilis, cholera, chlamydiasis, genital warts, Diabetes, Hepatitis-B, Genital Herpes, AIDS, Gonorrhoea, malaria :

(1) 3

(2) 4

(3) 7

(4) 5

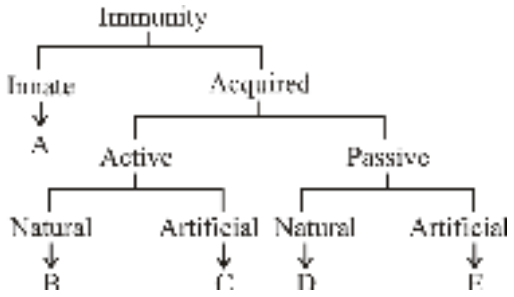
40)

Match the columns I and II, and choose the correct combination from the options given :-

	Column-I		Column-II
(a)	Amoebiasis	(i)	Treponema pallidum
(b)	Diphtheria	(ii)	Use only sterilised food and water
(c)	Cholera	(iii)	DPT vaccine
(d)	Syphilis	(iv)	Use oral rehydration therapy

(1) a-i, b-ii, c-iii, d-iv
(2) a-ii, b-iii, c-iv, d-i
(3) a-i, b-ii, c-iv, d-iii
(4) a-ii, b-iv, c-i, d-iii

41) Study the given flow chart and identify A, B, C, D, E.



	A	B	C	D	E
(1)	Clinical infection	Interferons	vaccine	Colostrum	Antiserum
(2)	Interferons	Clinical infection	vaccine	Colostrum	Antiserum
(3)	Interferons	Clinical infection	Colostrum	vaccine	Antiserum
(4)	Interferons	vaccine	Clinical infection	Sera	Antiserum

(1) 1
(2) 2
(3) 3
(4) 4

42) Given below are two statements:
Statement I: Cellular oncogenes (c-onc) or proto-oncogenes, when activated under certain conditions, can lead to the oncogenic transformation of the cells.
Statement II: Cancer detection techniques such as biopsy and histopathological studies are not

useful for detecting cancers of the internal organs.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II are false

(4) Statement I is true but Statement II is false

43) Assertion A: Pneumonia can lead to discoloration of lips and fingernails, turning them gray to bluish.
Reason R: In pneumonia, the alveoli are filled with fluid, which hampers the oxygen exchange in the lungs.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

44) Given below are two statements:
Statement I: Many fungi belonging to the genera Microsporum, Trichophyton, and Epidermophyton are responsible for ringworms, which is one of the most common infectious diseases in humans.
Statement II: Ringworms are generally acquired only by using towels, clothes, or the comb of infected individuals and not from soil.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true.

(2) Both Statement I and Statement II are true.

(3) Both Statement I and Statement II are false.

(4) Statement I is true but Statement II is false.

45) Given below are two statements:
Statement I: Vaccines work by introducing a weakened or inactivated pathogen into the body, which stimulates the production of antibodies and the generation of memory B and T-cells.
Statement II: In cases where a quick immune response is required, such as with tetanus, injecting preformed antibodies or antitoxin is more effective than vaccination.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true.

(2) Both Statement I and Statement II are true.

(3) Both Statement I and Statement II are false.

(4) Statement I is true but Statement II is false.

46) Myotonic Dystrophy is which type of disease-

(1) Autosomal recessive

(2) Autosomal dominant

(3) X-linked recessive

(4) X-linked dominant

47) Which of the following is not chromosomal disorders :

(1) Down's syndrome
(2) Kline filter's syndrome
(3) Phenyl ketonuria
(4) Turner's syndrome

48) Match the following. **[NCERT, (XII) Pg. # 88]**

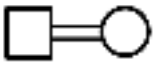
(i)		(A)	Monozygotic twins
(ii)		(B)	Carrier for sex linked recessive disease
(iii)		(C)	Carrier for autosomal recessive disease
(iv)		(D)	Normal female

(1) (i) – A, (ii) – B, (iii) – C, (iv) – D
(2) (i) – D, (ii) – C, (iii) – A, (iv) – B
(3) (i) – C, (ii) – D, (iii) – B, (iv) – A
(4) (i) – B, (ii) – C, (iii) – D, (iv) – A

49)

Which of the following symbol represent consaguineous marriage.

(1) 

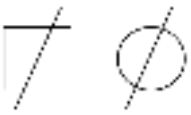
(2) 

(3) 

(4) 

50) Match the following columns.

Column - I		Column - II	
A.		1.	Death

B.		2.	Five unaffected offsping
C.		3.	Sex unspecified
D.		4.	Female
E.		5.	Male

Codes:

	A	B	C	D	E
(1)	1	2	5	4	3
(2)	1	2	4	3	5
(3)	1	2	3	4	5
(4)	5	4	3	2	1

(1) 1

(2) 2

(3) 3

(4) 4

51) β-thalassemia are due to mutation in the -

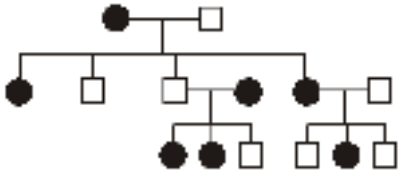
(1) HBA 1 gene on 16th chromosome

(2) HBA 2 gene on 16th chromosome

(3) HBA 1 gene on 11th chromosome

(4) HBB gene on 11th chromosome

52) Given pedigree show inheritance of autosomal dominant disorder (Myotonic dystrophy) then.



What will be genotype of shaded symbol :

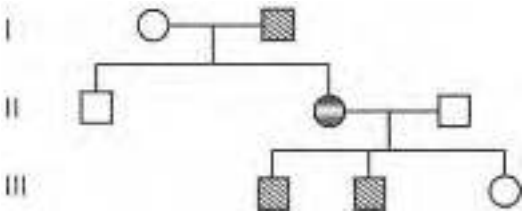
(1) AA

(2) Aa

(3) AA or Aa

(4) aa

53) Analyse the following pedigree and select the correct option.



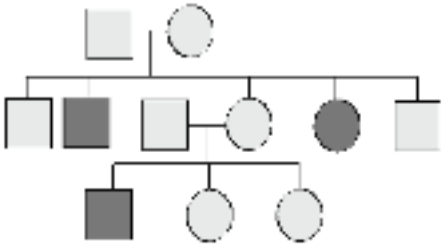
(1) The given pedigree is applicable for Edward's syndrome

(2) The genotype of female in first generation can be XX.

(3) The genotype of female in third generation is X^cX

(4) Male parent of generation I is suffering from Klinefelter's syndrome

54) Given pedigree chart shows inheritance of autosomal recessive trait (for eg - sickle cell anaemia) then what will be genotype of parents?

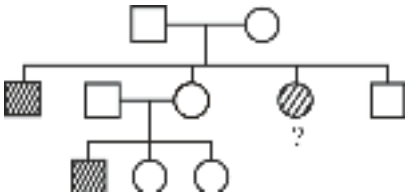


(1) Father - (Aa) Mother - (aa)

(2) Father - (aa) Mother - (aa)

(3) Father - (Aa) Mother - (Aa)

(4) Father - (AA) Mother - (AA)



55) What should be the genotype of the indicated member ?

(1) AA

(2) Aa

(3) XY

(4) aa

56) Sickle cell-anaemia disorder arises due to

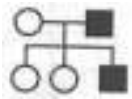
(1) Duplication of a segment of DNA

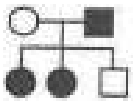
(2) Substitution in a single base of DNA

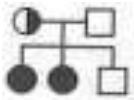
(3) Deletion of a segment of DNA

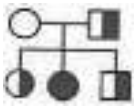
(4) Duplication in a base pair of RNA

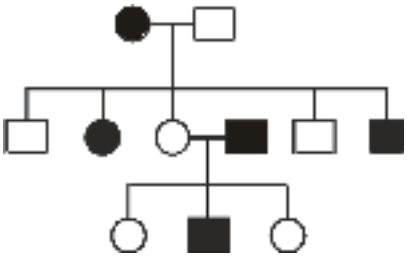
57) Wife is PTC non-taster and husband is PTC taster. Their son is taster but daughters are non-tasters. This is not a sex linked trait. Which pedigree is correct?

(1) 

(2) 

(3) 

(4) 

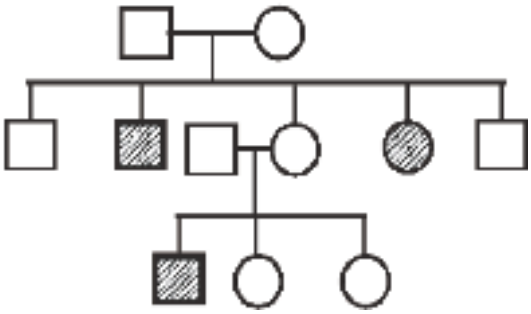
58) Study the pedigree chart of a family given below:  it is representative for disease like :

(1) Sickle cell anaemia, Haemophelia

(2) Myotonic dystrophy, Thalassemia

(3) Phenylketonuria, Colour blindness

(4) Pseudoricketes, sickle cell anaemia

59) 

In this given pedigree what is the mode of inheritance

(1) Autosomal dominant

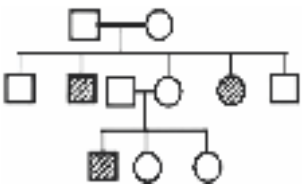
(2) Autosomal recessive

(3) X-linked dominant

(4) X-linked recessive

60)

In this given pedigree what is the mode of inheritance :



(1) Autosomal dominant
(2) Autosomal recessive
(3) X-linked dominant
(4) X-linked recessive

61) A classical example of point mutation is

(1) Chronic myelogenous leukemia
(2) Turner’s syndrome
(3) Sickle cell anaemia
(4) Down’s syndrome

62) The exchange of chromosomal parts between non homologous pairs of chromosomes is :

(1) Crossing over / Transduction
(2) Translocation
(3) Inversion
(4) Transition

63) Type of gene mutation which involves replacement of purine with pyrimidine or vice versa (OR) The substitution of one type of base with another type of base is

(1) Transduction
(2) Transversion
(3) Translocation
(4) Transcription

64) Chromosome number 2n-1 is an example of

(1) Trisomy
(2) Euploidy
(3) Polyploidy
(4) Monosomy

65) Which of the following condition in human beings is incorrectly matched ?

(1) Down syndrome – Autosomal trisomy
(2) Turner syndrome – Autosomal monosomy
(3) Klinfelter syndrome – Sex-chromosomal trisomy
(4) Cry-du chat syndrome – Chromosomal deletion

66) Which of following is an example of transversion mutation?

(1) A replaced by T
(2) A replaced by G
(3) C replaced by T

(4) Both (2) and (3)

67) **Assertion (A) :** Mutation is a phenomenon which results in alteration of DNA sequences.
Reason (R) : Recombination and mutation leads to variation in DNA.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

68) **Assertion :** Sickle cell anemia is a classical example of point mutation.
Reason : Sickle cell anemia arises due to change in single base pair of DNA.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

69) Under which of the following conditions will there be **no** change in the reading frame of following mRNA ?
5' UAC AGC GGU GCU AUA 3'

(1) Insertion of A at 5th position
(2) Insertion of A and G at 4th and 5th position respectively.
(3) Deletion of GGU from 7th, 8th and 9th positions respectively.
(4) Deletion of G from 5th position

70) Sickle cell anaemia occurs due to change in amino acids from _____:

(1) Valine to Glutamate
(2) Glutamate to Valine
(3) Glutamine to Valine
(4) Valine to Glutamine

71) Match the columns

Column – I	Column - II
a. Down’s syndrome	p. Presence an extra sex chromosome
b. Cri-du-chat	q. Loss of short arm of chromosome 5
c. Klinefelter’s syndrome	r. Absence of a sex chromosome
d. Turner’s syndrome	s. Presence of an extra somatic chromosome

(1) a-s, b-q, c-p, d-r
(2) a-r, b-s, c-p, d-q
(3) a-s, b-p, c-q, d-r
(4) a-s, b-q, c-r, d-p

72) Given below are two statements.
Statement-A : Frame shift mutation (Gibberish mutation) provides proof that genetic code is commaless.
Statement-B : Frame shift mutation involves the change the protein product due to addition or deletion.
Answer from the options given below.

(1) Statement A is incorrect but statement B is correct

(2) Statement A is correct but statement B is incorrect

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

73) The mutation seen in the below given representation is



(1) Substitution mutation

(2) Chromosomal aberration

(3) Frame-shift mutation

(4) Deletion mutation

74) Transition type of gene mutation is caused when : **(Replica of Q.129)**

(1) GC is replaced by TA

(2) CG is replaced by GC

(3) AT is replaced by CG

(4) AT is replaced by GC

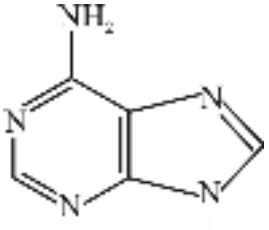
75) Observe the base sequence given below:
CCACAAGTTGACCGA
123456789101112131415
Which of the following will not lead to frame shift mutation?

(1) Insertion of the base 'G' after 5th base

(2) Deletion of the 5th base

(3) Deletion of 4th and 5th base together

(4) Deletion of 7th, 8th and 9th base together



76) What is the correct about given chemical compound :- (i) It can be present in both RNA and DNA.
(ii) It is purine type N-base.
(iii) It is nucleoside.

(iv) It is cytosine.

(1) Only iii and iv
(2) Only i and ii
(3) Only i and iii
(4) Only ii and iv

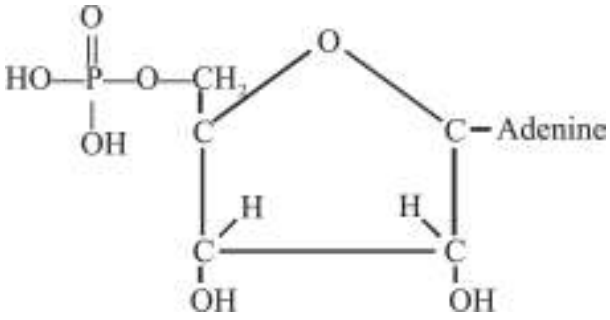
77) Thymine is :-

(1) A Nucleoside
(2) 5 – Methyl uracil
(3) 4 – Methyl uracil
(4) A Nucleotide

78) Phosphate residue is absent in :-

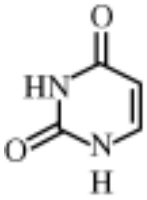
(1) Adenylic Acid
(2) Cephalin
(3) Uridine
(4) Cytidylate

79) Following structure is :-



(1) Deoxyribonucleotide
(2) Ribonucleotide
(3) Deoxyribonucleoside
(4) Ribonucleoside

80) Identify the following molecule shown below with their nature and occurrence from the given table.



	Molecule	Nature	Occurrence
(1)	Uracil	Nucleotide	Found in RNA
(2)	Uridine	Nucleoside	Found in both RNA and DNA
(3)	Uracil	Nitrogen base	Found in RNA

(4)	Uridine	Nucleoside	Found in t-RNA

(1) 1

(2) 2

(3) 3

(4) 4

81) Thymidine contains :-

(a) Thymine (b) Ribose

(c) Deoxyribose (d) Purine base

(e) Pyrimidine base

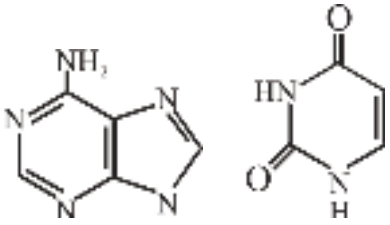
Options

(1) a, b and e

(2) a, c and d

(3) a, b and d

(4) a, c and e

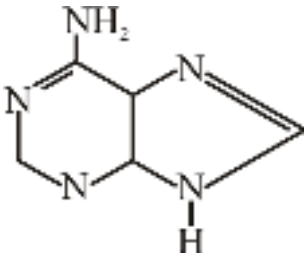
82)  The diagrams represent the nitrogenous bases. Identify the correct combination :-

(1) A → Adenine; B → Thymine

(2) A → Guanine; B → Uracil

(3) A → Adenine; B → Uracil

(4) A → Guanine; B → Thymine

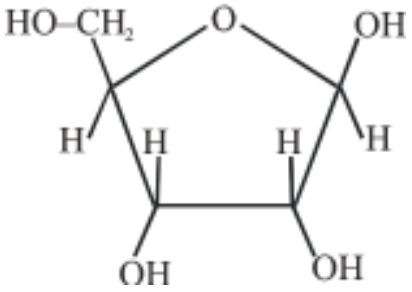
83) The following structure is related to which compound ? 

(1) Cytosine

(2) Uracil

(3) Adenine

(4) Thymine

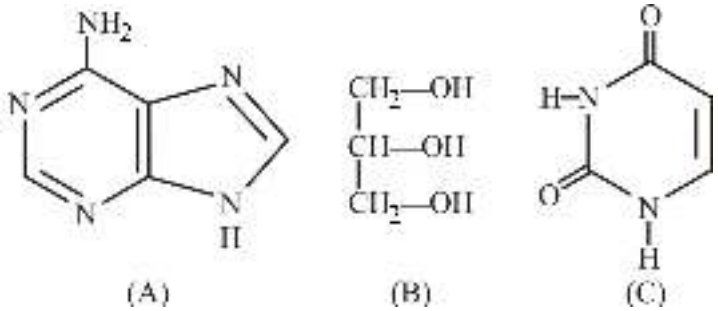
84) 

(1) It is structure of deoxyribose

(2) It is component of deoxyribonucleotide

(3) It is non reducing monosaccharide

(4) It is present in RNA

85) 

Which of the above may be present in both RNA & DNA ?

(1) A

(2) B

(3) C

(4) Both A & B

86) What is the major hallmark of double helix modes of DNA given by Watson and Crick :-

(1) Base pairing between two stands.

(2) Presense of phosphodiester bond.

(3) Acidic nature of DNA.

(4) Presence of pentose sugar.

87) In Brassica Campestris [mustard] a DNA having 4.8 kilo bases was found. And on analyzing it, 800 adenine molecules were found then how many cytosine molecules will be present in the DNA sample ?

(1) 800

(2) 4000

(3) 1600

(4) 3200

88) The low melting area of DNA is rich in :-

(1) A-T base pairs

(2) A-U base pairs

(3) G-C base pairs

(4) Both (1) and (2)

89) Negative charge of DNA is due to which of the following constituent

(1) Sugar

(2) Nitrogenous base

(3) Phosphoric acid

(4) Hydroxyl group (–OH) present on sugar

90) If a highly purified sample of plant DNA is dissolved in distilled water, the resulting solution will be ?

(1) Neutral

(2) Basic

(3) Acidic

(4) Highly basic

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	2	3	3	3	3	4	1	1	3	3	3	2	4	1	4	3	3	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	3	4	3	1	3	1	3	4	2	2	3	1	4	2	3	4	2	1
Q.	41	42	43	44	45															
A.	3	3	4	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	2	3	1	3	4	1	3	1	2	1	3	3	4	1	3	4	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	4	1	1	1	1	1	4	3	2	4	2	2	2	3	2	4	1	3
Q.	86	87	88	89	90															
A.	4	2	2	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	3	2	1	3	3	1	3	4	3	4	3	1	3	3	1	1	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	2	3	3	2	4	2	2	3	1	3	4	2	2	3	1	4	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	1	4	2	2	3	2	2	4	4	2	3	3	4	2	1	2	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	2	4	3	1	2	1	3	2	1	3	1	4	4	2	2	3	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	3	3	4	1	1	3	1	3	3										

SOLUTIONS

PHYSICS

1) case (ii) contradicts Gauss' law for magnetic fields

2) $B_{axial} = \frac{\frac{\mu_0}{4\pi}\frac{2Mr}{(r^2-\ell^2)^2}}{}$ {r is the distance from the centre Magnetic moment m = 2m × l}
At center r = 0
$$B = \frac{\mu_0 2\mu(0)}{4\pi(0-\ell^2)^2} = 0$$

3) $2B_0L^2Wb$

4) $B = \frac{\mu_0}{4\pi}\frac{M}{r^3}\sqrt{1+3\cos^2\theta}$
$$= \frac{\mu_0 M}{4\pi r^3}\sqrt{1+3\times\frac{3}{4}} = \frac{\mu_0 M}{4\pi r^3}\times\sqrt{\frac{13}{4}}$$
$$= \frac{4\pi\times10^{-7}\times10}{4\pi\times10^{-3}}\times\sqrt{\frac{13}{4}}$$
$$B = 10^{-3}\times\sqrt{\frac{13}{4}}T$$

5) $I = \int J.ds$
$$= \int_0^R J_0\left(1-\frac{x}{R}\right).2\pi x dx$$
$$= 2\pi J_0\int_0^R\left(x-\frac{x^2}{R}\right)dx$$
$$I = 2\pi J_0\left(\frac{R^2}{2}-\frac{R^2}{3}\right)$$
$$I = 2\pi J_0\frac{R^2}{6}$$
$$I = \frac{\pi J_0 R^2}{3}$$

6)

Generated by Allie

Problem Statement: The question asks what properties the core material of an

electromagnet must have. Specifically, it wants to know which combination of retentivity and permeability is ideal for the ferromagnetic material used in electromagnets.

Underlying Concept: Electromagnets rely on magnetic properties of materials to create strong magnetic fields when electric current flows through a coil wrapped around the core. Two key properties of core materials are permeability and retentivity. Permeability measures how easily a material can be magnetized or penetrated by magnetic field lines, while retentivity measures how well the material retains magnetism after the external magnetizing force (like current) is removed. For an electromagnet, the core should magnetize easily (high permeability) and lose magnetism quickly when the current stops (low retentivity) to allow switching the magnetic field on and off effectively.

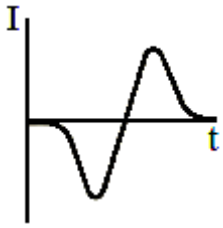
Tips and Tricks: Remember: - High permeability means strong response to magnetic fields. - Low retentivity means minimal leftover magnetism, allowing the magnet to quickly demagnetize. A good way to recall this is that electromagnets need to be 'responsive but forgetful.'

Common Mistakes: Confusing retentivity with permeability and assuming both need to be high. Many students think that high retentivity is always desirable, but for electromagnets, high retentivity causes residual magnetism, which is undesirable.

Why Other Options Are Incorrect?: Options suggesting high retentivity are incorrect because such materials retain magnetism after current stops, making the magnetization difficult to switch off. Options with low permeability are wrong because the core won't magnetize easily, resulting in a weak magnetic field.

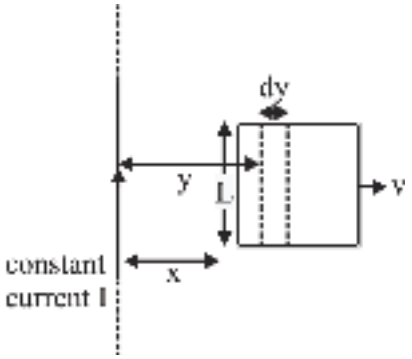
7) Induced electric field in copper ring is more in magnitude than wooden ring.

8) **Sol :**



9) $(0.05)\left(\hat{i}-\sqrt{3}\hat{k}\right) A-m^2$

10) Consider a small segment of width dy at a distance y from the wire. Let flux through the segment be dϕ



$$\therefore d\phi = \frac{\mu_0 I}{2\pi y} L\, dy$$

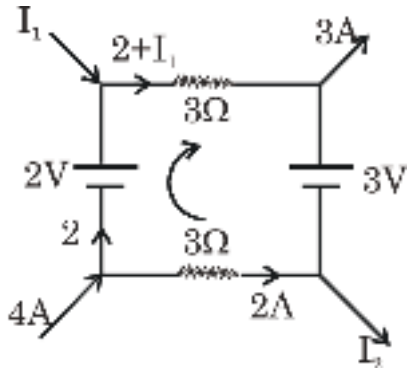
$$\therefore \phi = \frac{\mu_0 IL}{2\pi}\int_x^{x+b}\frac{dy}{y}$$

$$= \frac{\mu_0 IL}{2\pi}[\ell n(x+b) - \ell n\ x]$$

Now $\frac{d\phi}{dt} = \frac{\mu_0 IL}{2\pi}\left[\frac{1}{x+b}\frac{dx}{dt} - \frac{1}{x}\frac{dx}{dt}\right]$

$$= \frac{\mu_0 IL}{2\pi}\left[\frac{(-b)}{x(x+b)}\right] v = \frac{-\mu_0 IbLv}{2\pi x(x+b)}$$

$\therefore$ Induced emf = $= \frac{\mu_0 IbLv}{2\pi x(x+b)}$



11)

Apply KCL

$I_1 + 4 = 3 + I_2$

$I_1 + 1 = I_2$ ... (i)

KVL in loop

$2 - 3(2 + I_1) - 3 + 6 = 0$

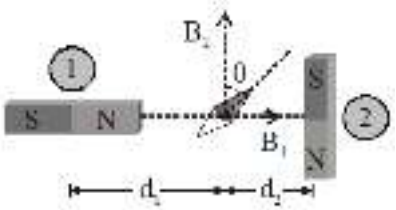
$2 - 6 - 3I_1 + 3 = 0$ ...(ii)

$I_1 = -\frac{1}{3}$ A

$I_2 = \frac{2}{3}$

12)

In equilibrium $B_1 = B_2 \tan\theta$



$$\Rightarrow \frac{\mu_0}{4\pi}.\frac{2M}{d_1^3} = \frac{\mu_0}{4\pi}.\frac{M}{d_2^3}\tan\theta$$

$$\Rightarrow \frac{d_1}{d_2} = (2\cot\theta)^{1/3}$$

13) $B_{eq} = \frac{K\sqrt{3}M}{r^3} = B$

After rotation $M_{net} = M$ ($\theta = 120°$)

$$B' = \frac{KM}{r^3} = \frac{B}{\sqrt{3}}$$

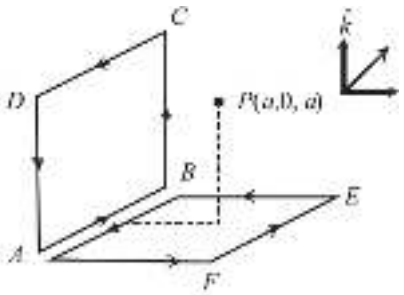
14) $M_1 = \frac{M}{3}$ and $M_2 = \frac{2M}{3}$

$$M_{net} = \sqrt{M_2^2 + M_2^2}$$

15) The magnetic field at due to the loop is equal to the vector sum of the magnetic fields produced by loops ABCDA and AFEBA as shown in the figure.

Magnetic field due to loop ABCDA will be along $\hat{i}$ and due to loop AFEBA, along $\hat{k}$. Magnitude of magnetic field due to both the loops will be equal.

Therefore, direction of resultant magnetic field at P will be $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$.



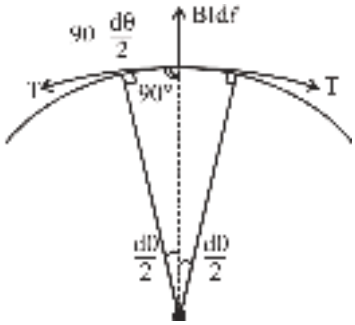
16) $d_{cm} = \frac{l}{2} + \frac{l}{4} = \frac{3l}{4}$

$$I = \frac{Ml^2}{12} + \frac{Ml^2}{3} + Ml^2 = \frac{17Ml^2}{12}$$

$$I = \frac{2\pi}{3}\sqrt{\frac{17l}{2g}}$$

17) If bar magnet is falling vertically through the hollow region of long vertical copper tube, the magnetic flux linked with the copper tube (because of 'non-uniform' magnetic field of

magnet) changes and eddy currents are generated in the body of the tube.
By Lenz's law the eddy currents opposes the falling of the magnet which thus experiences a retarding force. The retarding force increases with increasing velocity of the magnet and finally equals the weight of the magnet. The magnet then attains a constant final terminal velocity means magnet ultimately falls with zero acceleration in the tube.



18)

$$\frac{d\ell}{R} = d\theta$$

$$\Rightarrow \quad d\ell = Rd\theta$$

$$2T\cos\left(90° - \frac{d\theta}{2}\right) = BIdl$$

$$\Rightarrow \quad 2T\sin\frac{d\theta}{2} = BId\ell$$

$$\Rightarrow \quad 2T.\ \frac{d\theta}{2} = BIRd\theta$$

(as dθ is very small $\sin\frac{d\theta}{2} = \frac{d\theta}{2}$)

$$\Rightarrow \quad T = BIR = BI\frac{L}{2\pi} \qquad (\because L = 2\pi R)$$

19) By using heat loss = VA ; where V = volume = 10^{-3} m^3; A = Area = 0.1m^2,

Heat loss = $10^{-3} \times 0.1$

$= 10^{-4}$ J

20)

$\tau = K\theta$

$NiAB = K\theta$

$$A = \frac{K\theta}{NiB} = \frac{4 \times 10^{-5} \times 0.05}{200 \times 10 \times 10^{-3} \times 0.01}$$

On solving A = 10^{-4} m^2 = 1 cm^2

21) Distance travelled by the loop in t =2s
= (1cm/sec) × 2 sec = 2cm
So that flux of the loop is increasing
e = Bvℓ = 0.6 × 1 × 10^{-2} × 5 × 10^{-2}

$= 3 \times 10^{-4}$ V
At t = 10s loop will be completely inside in the magnetic field B
ϕ = contant $\Rightarrow$ e = 0 (No emi)

22) **Problem Explanation:**
Two bulbs, 60 W (at 110 V) and 100 W (at 110 V), are connected in series to a 220 V supply. We need to determine which bulb will fuse.
Concept:
- In a series circuit, the same current flows through both bulbs.
- If a bulb receives voltage beyond its rating, it will fuse.

Formula:
- Resistance of a bulb:
$$R = \frac{v^2}{P_x}$$
- Total resistance in series:
$$R_{total} = R_{60} + R_{100}$$
- Current in the circuit:
$$I = \frac{V_{total}}{R_{total}}$$
- Voltage across each bulb:
$$V = I \times R$$

Calculation:
- Resistance of bulbs:
$$R_{60} = \frac{110^2}{60} = 202\,\Omega,\ R_{100} = \frac{110^2}{100} = 121\,\Omega,$$
- Total resistance in series:
$$R_{total} = 202 + 121 = 323\ \Omega$$
- Current in the circuit:
$$I = \frac{220}{323} \approx 0.68\,A$$
- Voltage across each bulb:
$$V_{60} = 0.68 \times 202 = 137.4V,\ V_{100} = 0.68 \times 121 = 82.3V$$

Conclusion:
- The 60 W bulb receives 137.4 V, which is above its 110 V rating, so it will fuse.
- The 100 W bulb receives 82.3 V, which is below its rating, so it remains safe.

Final Answer:
60 W bulb
Hence, option (1) is correct.

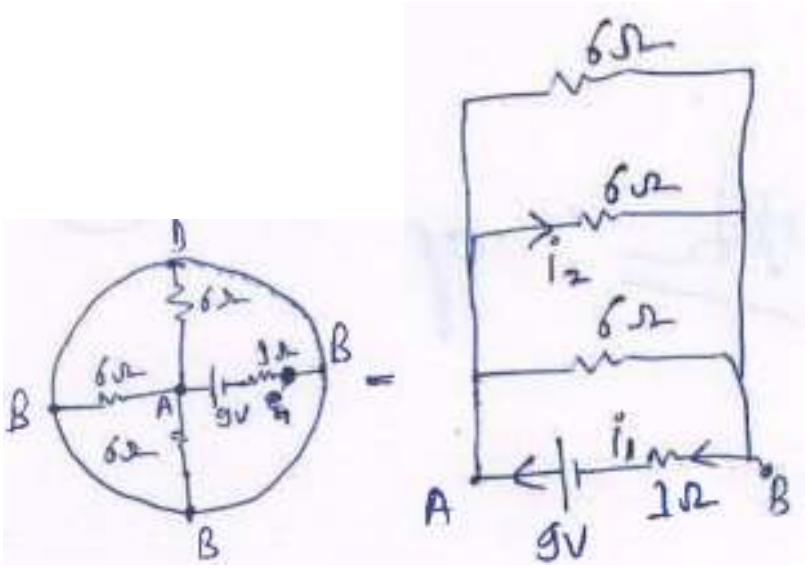
23) **1. Question Explain:**
Find the current readings on ammeters A_1 and A_2 in a circuit with resistors and a 9V battery using Ohm's Law and Kirchhoff's Laws.
2. Concept Based:
Use Ohm's Law V = IR and current division for resistors in parallel. Analyze series-parallel connections and current distribution across branches.

3. Formula Used:

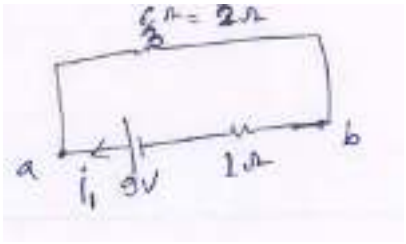
A. Total resistance in series: $R = R_1 + R_2$

B. Total resistance in parallel: $\frac{1}{R_{n_1}} = \frac{1}{R_1} + \frac{1}{R_2} \quad - \quad -$

C. Ohm's Law: $I = \frac{V}{R}$

Concept diagram: 

According to Diagram there are Three 6Ω resistance in Parallel given



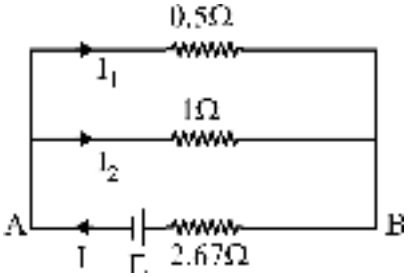
then i, $= \frac{9}{2+1} = 3A$

$V_{ab} = 9 - (i_1)(1\Omega)$

$= 9 - (3) \quad \Rightarrow 6V$

24)

Resistance of bulb, $R = \frac{(1.5)^2}{4.5} = 0.5\ \Omega$



Bulb glow with full intensity of $V_A - V_B = 1.5$ V

$I_1 = \frac{1.5}{0.5} = 3A$ & $I_2 = \frac{1.5}{1} = 1.5A$

So, $I = I_1 + I_2 = 4.5$ A

and $V_A - V_B = E - I(2.67)$

$1.5 = E - 4.5(2.67)$

$\Rightarrow E = (1.5 + 12)V = 13.5$ V

25) $R_1 = \tan\theta \quad R_2 = \tan(90-\theta) = \cot\theta$

$R_1 = R_0(1 + \alpha T_1) \quad R_2 = R_0(1 + \alpha T_2)$

$R_2 - R_1 = R_0 \alpha (T_2 - T_1)$

$\cot\theta - \tan\theta \alpha (T_2 - T_1)$

$$T_2 - T_1 \alpha \frac{2(\cos^2\theta - \sin^2\theta)}{2\cos\theta\sin\theta}$$

$$T_2 - T_1 \alpha \frac{2\cos 2\theta}{\sin 2\theta} \propto \cot(2\theta)$$

26) Temperature coefficient of resistance is defined as the fractional change in resistance per unit change in temperature.

$$\alpha = \frac{1}{R}\cdot\frac{dR}{dt} = \frac{1}{R_0(1 + at + bt^2)}[R_0(a + 2bt)]$$

$$= \left(\frac{a + 2bt}{1 + at + bt^2}\right)$$

27) $I_g = \frac{5}{4900 + 100} \propto 40$;

$I_g' \propto \frac{5}{4900 + 100 + R} \propto 20$

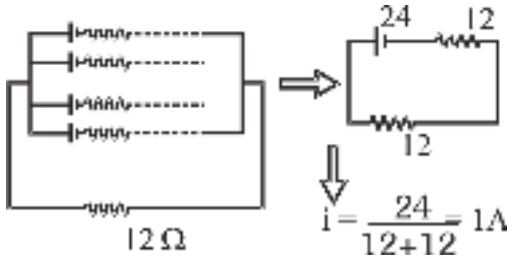
When total resistance is (100 + 4900) = 5000 Ω

current = 40 div to reduce this up to 20 div total resistance must be 10000Ω resistance = 5000 Ω

$$\frac{\frac{5}{5000}}{\frac{5}{5000+R}} = \frac{40}{20} = 2$$

$$\frac{5000 + R}{5000} = 2 \Rightarrow \boxed{R = 5000\Omega}$$

28)



For $I_{max} = \frac{nr}{m} = R$ ...(1)

n = number of coloum

m = number of row

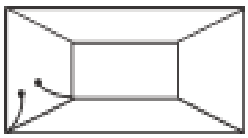
nm = 48 ...(2)

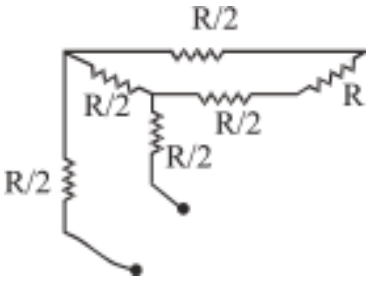
from eq (1) and (2)

n = 12

m = 4

29)

≡ 



≡

use series & parallel

$$= \frac{2R \times \frac{R}{2}}{2R + \frac{R}{2}} + \frac{R}{2} + \frac{R}{2} = \frac{2R}{5} + R = \frac{7R}{5}$$

30)

Generated by Allie

Problem Statement: The question asks to find the electrical power developed in a circular loop of radius r and resistance R when a time-varying magnetic field B = B_0 e^(-t) is applied inside the loop and the switch connected to the loop is closed.

Underlying Concept: The problem involves electromagnetic induction, specifically Faraday's law of induction, which states that a time-varying magnetic flux through a loop induces an electromotive force (emf) in the loop. The induced emf causes current to flow through the resistance, producing electrical power.

Relevant Formulas: Magnetic flux through the loop is given by $\phi = BA = B\pi r^2$ Induced emf is $E = -\frac{d\phi}{dt}$ Power dissipated in resistance R by current due to emf E is $P = \frac{E^2}{R}$ Where $B = B_0 e^{-t}$, r is the radius, and R is the resistance.

Step-by-Step Calculation: Step 1: Calculate magnetic flux: $\phi = \pi r^2 B_0 e^{-t}$ Step 2: Find induced emf using differentiation: $E = -\frac{d\phi}{dt} = -\pi r^2 \frac{d}{dt}(B_0 e^{-t}) = \pi r^2 B_0 e^{-t}$ Step 3: Evaluate emf right after closing switch (t=0): $E_0 = \pi r^2 B_0$ Step 4: Calculate power dissipated using $P = \frac{E_0^2}{R} = \frac{(\pi r^2 B_0)^2}{R} = \frac{\pi^2 r^4 B_0^2}{R}$

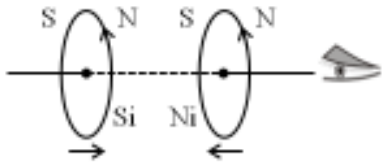
Tips and Tricks: Keep track of the negative sign in Faraday's law but note that power

depends on the square of emf, which makes it positive. Also, the area of the loop is πr^2, which is essential to calculate flux.

Common Mistakes: Forgetting to differentiate the exponential term correctly or ignoring the negative sign. Another mistake is calculating power using just emf without squaring it or dividing by resistance.

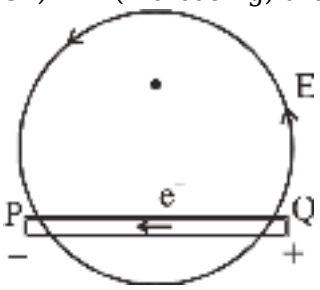
Why Other Options Are Incorrect?: Options that do not square the magnetic field term or the area, or that incorrectly include extra powers of r or constants, are incorrect. The correct answer must have B_0^2, π^2, and r^4 in the numerator divided by R.

31) Use lenz low



When loop moved towards each other then induced current in both are clockwise so that net current in both decrease.

32) $\otimes$B (increasing) then induced $\vec{E}$ = ACW



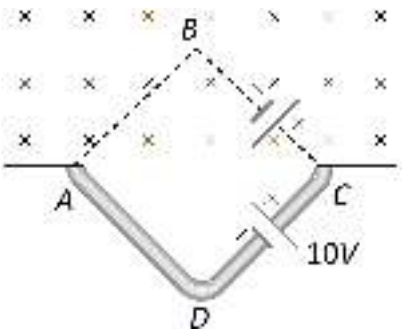
33) As the magnet falls, due to the variation of magnetic flux, induced current is developed in the hollow conductor so as to resist the cause of increment of flux; that means the velocity of the magnet is resisted. In other words we can say that the magnet is retarded before entering into the hollow conducting cylinder. Consequently its acceleration will be less than g because magnetic force opposes the gravitational force acting on the magnet. If the cylinder is too long, the magnetic force opposes the gravitational force on the magnet so as to yield zero acceleration after a long time to attain the terminal velocity. The eddy current in the cylinder results in its heating.

34) $\phi = 0$, e = 0

35) Given B = 0.01 - 2t Tesla ; $\frac{dB}{dt} = -2\text{Tesla/sec}$,
Induced emf

$$e = -\frac{d\phi}{dt} = -\frac{d}{dt}(BA) = -A\frac{dB}{dt} = -\frac{1}{2}(1^2) \times (-2)$$

$\Rightarrow$ e = 1V



Since magnetic field (×) decreasing so according to Lenz's law direction of induced current in upper part of square will be clockwise i.e., from A to C or in other words emf induces in a direction opposite to the main emf so resultant emf = 10 – 1 = 9V.

36) $R = r_1 - r_2$

37) is zero, other wise there would be a field falling as $\frac{1}{r^3}$ at large distances outside the toroid

38) domains are all perfectly aligned

39) $\frac{2}{3}$ Am^{-1}

40) The H field in the solenoid is (nearly) unchanged but the B field decreases drastically

41) may increase or decrease in size

42) a clockwise current is induced in loop A, but a counterclockwise current is induced in loop B

43) **Sol :** $i = \frac{B}{4\lambda}(b-a)$

44) **Sol :** 254

45) We know that electric field acts along high potential to low potential and direction of electric force is just opposite of the direction of electric field for electron.
Inside the plates no deviation occur in electron so

$f_m = f_E$

OR

If electron comes out from plate 3 and 4 then magnetic force $-e\left(\vec{V}\times\vec{B}\right)$ is upward means 4 to 3 initially, but in between it was $\left(-e\vec{E}\right)$ from plate 3 to 4, so direction of $\vec{E}$ is from 4 to 3. So plate 4 is at higher potential than 3.

If electron comes out from plate 1 and 2 then magnetic force $-e\left(\vec{V}\times\vec{B}\right)$ is downward means plate 1 to 2, but in between plates it was $\left(-e\vec{E}\right)$ from plate 2 to 1. So direction of $\vec{E}$ is from plate 1 to 2. So plate 1 is at higher potential than 2.

CHEMISTRY

46) A → $[Co(H_2O)_6]Cl_3$
B → $[Co(H2O)_5Cl]Cl_2.H_2O$
C → $[Co(H_2O)_4Cl_2]Cl.2H_2O$

47)

Co^{+3} → d^6, NH_3–SFL
t_{2g^6}
no unpaired electron

48) $[Pt(H_2O)_6]^{+4}$ $\Delta_0 > P$
with 4d/5d metal all ligand acts as SFL

49) $V^{+4}OSO_4$ unpaired electron is present in d-orbital.

50)

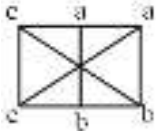
Question Explanation – The question asks you to identify which coordination complex(es) will have exactly three stereoisomeric forms.

Concept – The compound $[Cr(NO_3)_3(NH_3)_3]$ has two stereoisomers fac and mer-isomers. Both are optically inactive
The compound $K_3[Fe(C_2O_4)_3]$ has two stereoisomers. Both are optically active One is d from and the other is 1 form
The compound $[CoBrCl(en)_2]$ has three stereoisomers. Two are geometrical isomers cis and trans.
The cis isomer has two optical isomers ‘d’ from and ‘l’ from
Trans isomer is meso in nature
The compound $[CoCl_2(Ox)_2]^{3-}$ has three stereoisomers. Two are geometrical isomers cis and trans.
The cis isomer has two optical isomers ‘d’ from and ‘l’ from
Trans isomer is meso in nature

Final Answer : (1)

51) **Explaining the question :** Identify which of the listed cobalt complexes are chiral and thus exhibit optical isomerism.
Concept : Optical Isomerism **Solution :** Cis-$[Co(en)_2Cl_2]^+$ is chiral due to the adjacent

arrangement of ligands lacking a plane of symmetry. $[Co(en)_3]^{3+}$ is also chiral as the tris-chelated structure is non-superimposable on its mirror image.

52)  = Optically action

53) Octahedral Co(III) complexes with strong field ligands have zero magnetic moment.

54)

CFSE = – 0.8 Δ_0
CFSE = – 0.4 $\Delta_0 n_{t2g}$ + 0.6 Δ_0 n_{eg} as WFL
But t_{2g} = 2 CFSE = – 0.4 $\Delta_0(2)$ = –0.8 Δ_0
($t_{2g}^2 eg^0$) = 2 unp
$\mu = \sqrt{8}$
– 0.8 Δ_0 = – 0.4 $\Delta_0(5)$ + 0.06 $\Delta_0(2)$
= – 2Δ_0 + 1.2 Δ_0 = – 0.8 Δ_0 t_{2g}^5 eg^2
$\mu = \sqrt{15}$

55) MD13IC

56)

Generated by Allie

Problem Statement: The question asks which of the given complex ions is not expected to absorb visible light. In simpler terms, it wants us to identify the complex ion that does not show color due to electronic transitions in the d-orbitals when exposed to visible light.

Underlying Concept: The core concept is the electronic transitions in transition metal complexes, specifically d-d transitions. For a complex to absorb visible light via d-d transitions, it must have unpaired d-electrons so that electrons can be excited from one split d-orbital to another. Complex ions with all paired electrons (low-spin, strong field complexes) do not show such absorption and thus are usually colorless or do not absorb visible light.

Tips and Tricks: Remember that strong field ligands like CNâ» and NHâ‚ƒ tend to produce low-spin complexes with fewer or no unpaired electrons, while weak field ligands like Clâ» and Fâ» generally lead to high-spin complexes with unpaired electrons. Also, square planar geometry often causes pairing of electrons in dâ‚ complexes.

Common Mistakes: Students often overlook the effect of ligand field strength or assume all dâ‚ complexes will have unpaired electrons regardless of geometry. They might also forget that tetrahedral complexes are always high spin due to smaller splitting.

Why Other Options Are Incorrect?: 1. [NiClâ‚„]Â²â»: NiÂ²âº (dâ‚ˆ) in tetrahedral weak field ligand has unpaired electrons, so it absorbs visible light. 2. [Cr(NHâ‚ƒ)â‚†]Â³âº: CrÂ³âº

(dÂ³) has unpaired electrons regardless of field strength, so it absorbs visible light. 3. [FeFâ‚†]Â³â»: FeÂ³â¶ (dâµ) in weak field is high spin with unpaired electrons, so it absorbs visible light. Only [Ni(CN)â‚„]Â²â» with strong field CNâ» ligands in square planar geometry has all paired electrons, hence no d-d transitions and no visible absorption.

57) Refer configurations

58)

Cu has positive $E°(M^{2+}/M)$

59)

NCERT-XII, Pg # 223 & 236

60)

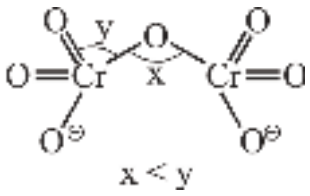
Conc.H_2SO_4 make explosive Mn_2O_7
→ HCl oxidises into Cl_2
→ HNO_3 it self is oxidising agent.

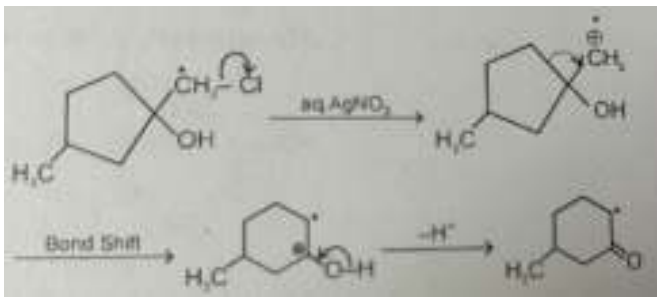
61)

$Cu_{(s)} + 4HNO_{3\,(aq)} \rightarrow Cu[NO_3]_{2\,(aq)} + 2NO_{2\,(g)} + 2H_2O_{(l)}$

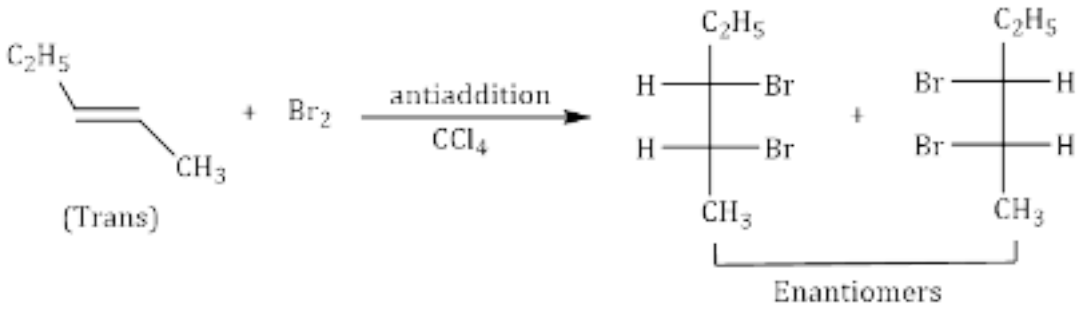
62)

Only Cl^- gives chromyl chloride test not covalent molecule

63) 

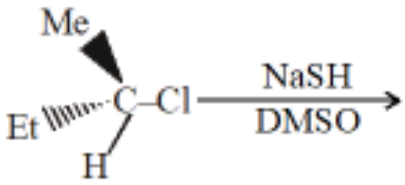
64) 

65) EAR of alkene.

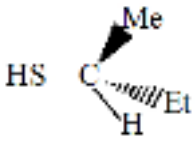


So, total no. of products = 2

66)



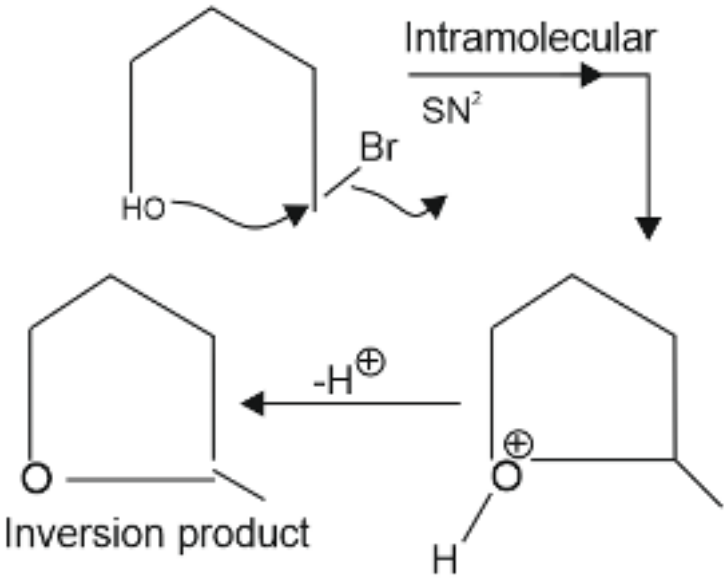
* SN^2 product will be formed
* Polar aprotic solvent is used, it favours the SN^2 Mechanism
* In SN^2 Nucleophile (here HS^-) attack from back side.
* Inversion of product will takes place

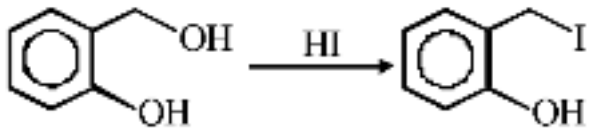


* New product
* A single stereo isomer will be formed
* Since product is altogether a different compound we cannot say about it dextro or levorotatory nature.

Only Option D is correct.

67)

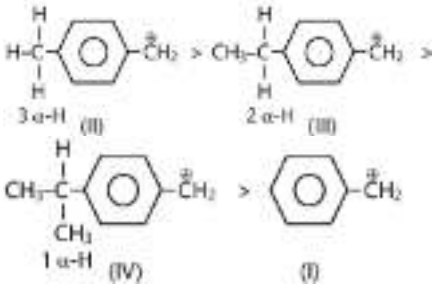


68) 

Stable bu R

69) (P) will form highly stabilised carbocation.
(R) will form highly destabilised carbocation, i.e., bridged carbocation.

70) Rate of hydrolysis by S_N1 ∝ stability of carbocation
In these, resonance is equal so we compare stability by hyperconjugation.

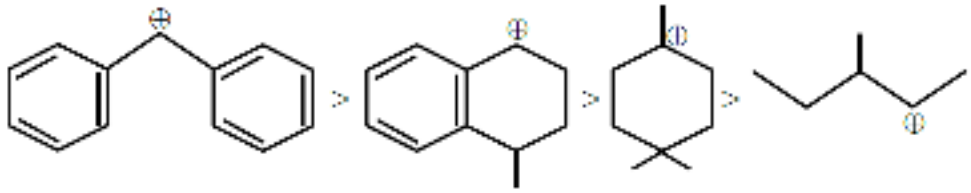


So, stability order: II > III > IV > I

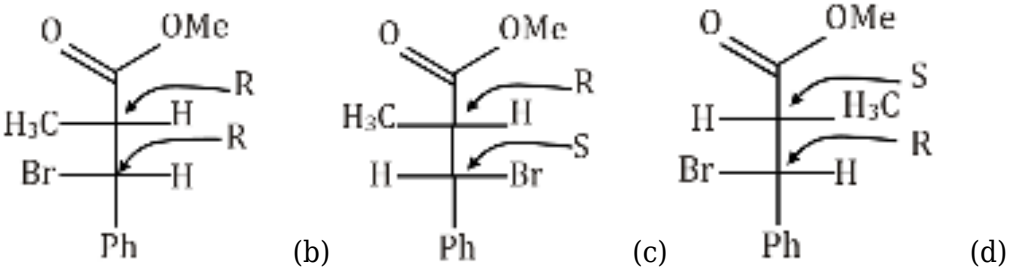
71)

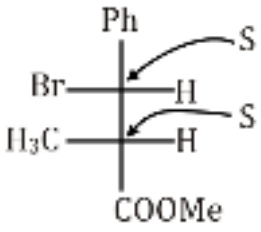
Solvolysis or S_N1 ∝ stability of carboccation

Stability order



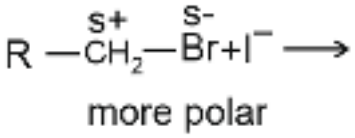
72) (A) In S_N2 these is complete inversion.

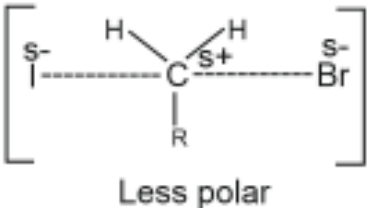




73) $R-CH_2Br + I^- R-CH_2-I + Br^-$

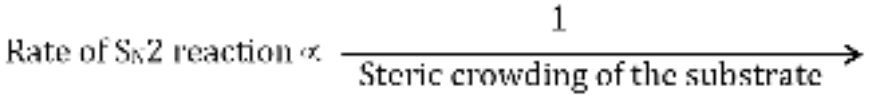
In SN^2 reactions, due to formation of transition state the magnitude of net charge decreases.



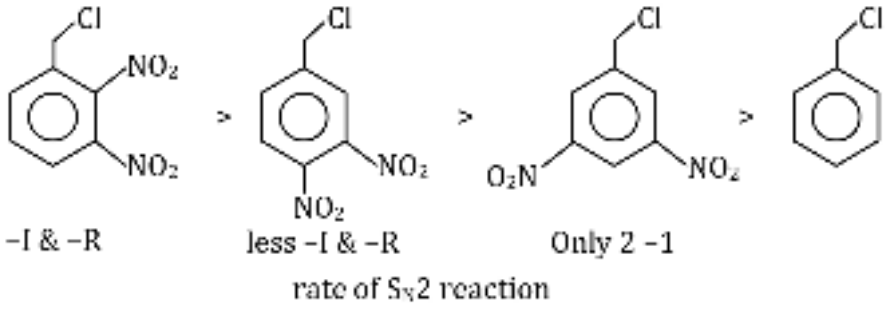


74)

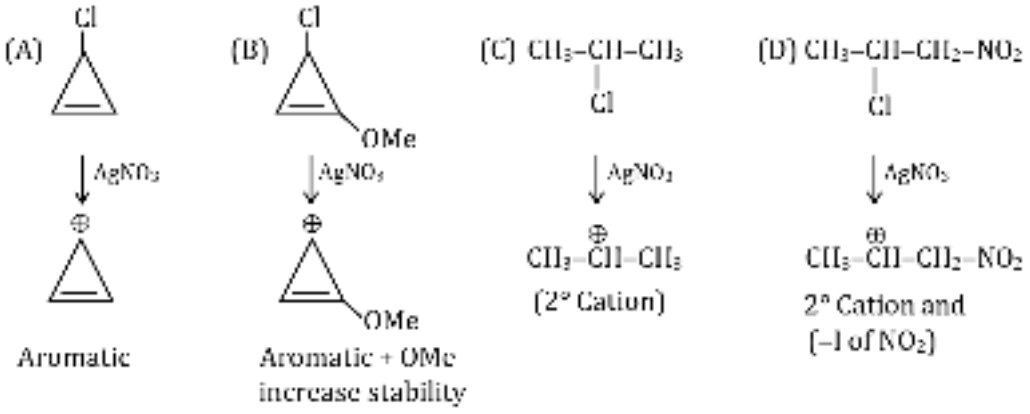
Generally, rate of S_N2 reaction ∝ Electrophilicity at the carbon where leaving group is attached.
So, Rate of S_N2 reaction ⇒
(D) (iii) > (ii) > (i)



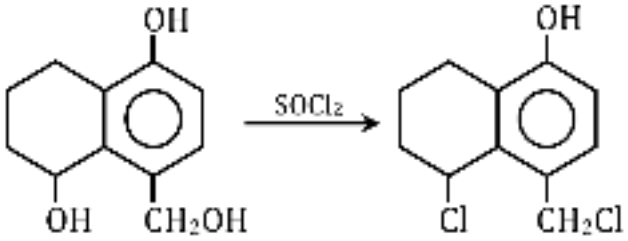
75)
So, rate of S_N2 reaction,
(3) (i) > (iii) > (ii)



76)

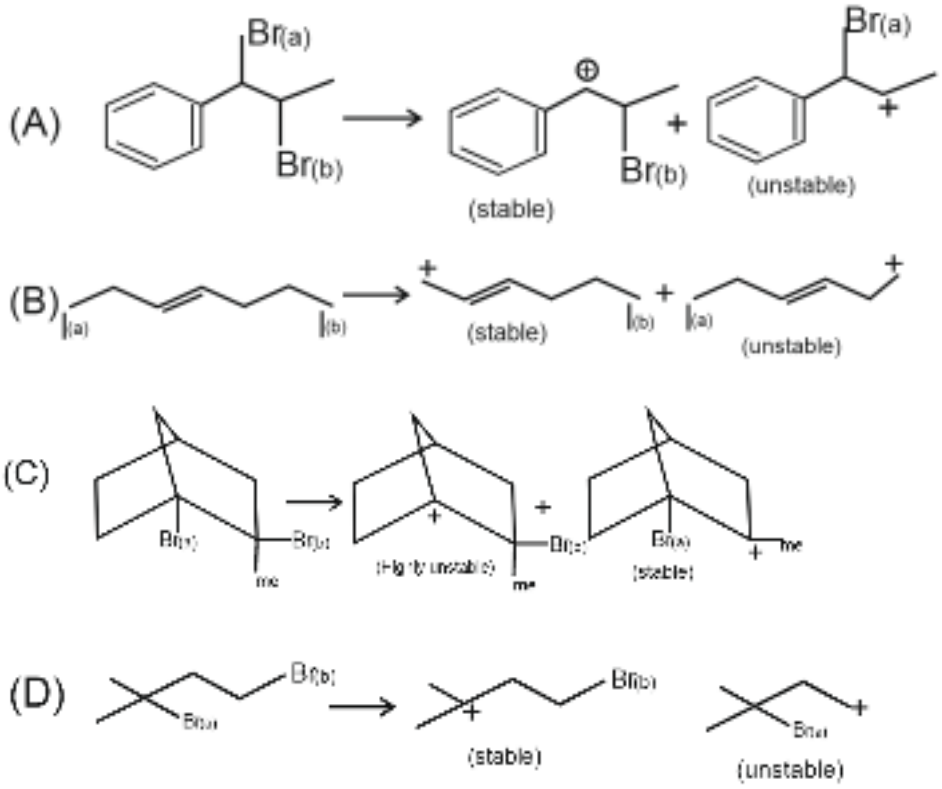


77)
 Stability Cation B > A > C > D

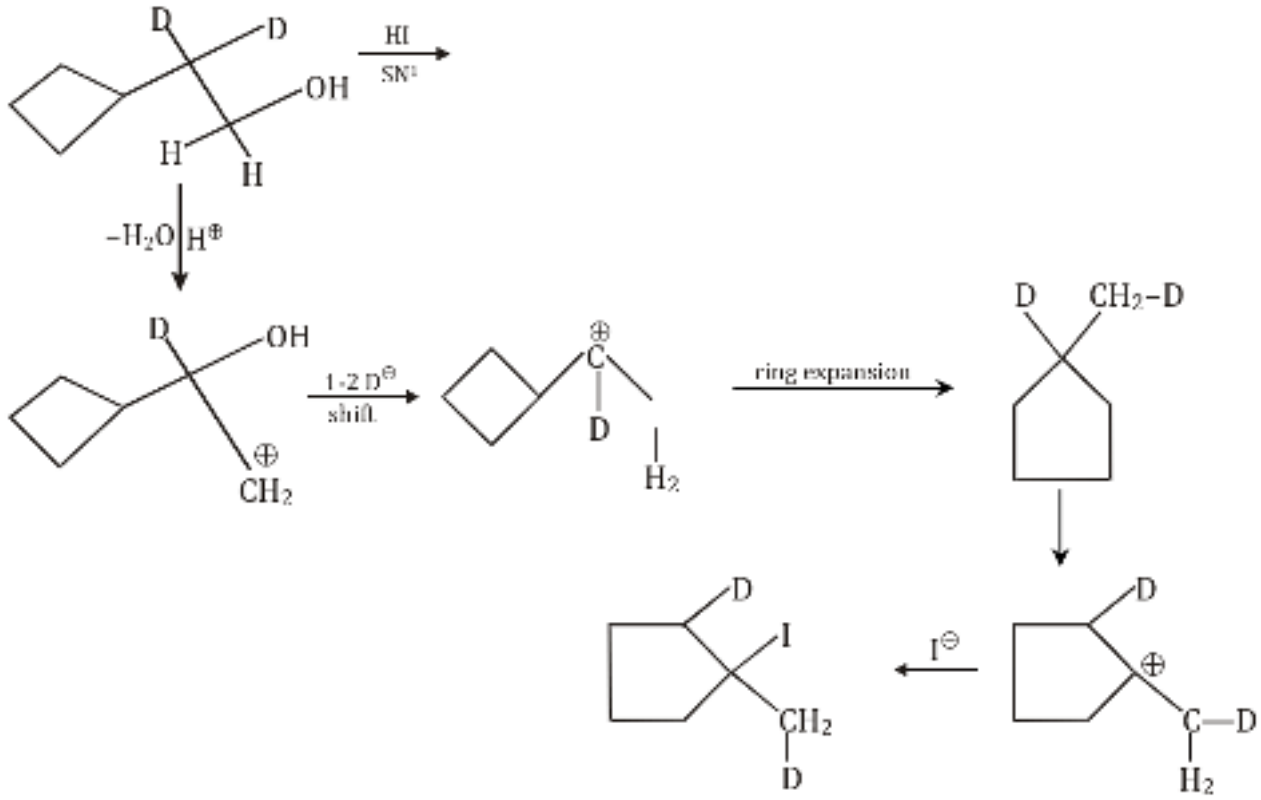
78) 

79)

Reactivity of SN^1 reaction α stability of carbocation



80)



81)

+ M group increases rate SN_1 reaction.

82) Reaction proceeds through S_N2 mechanism, so complete inversion product is obtained (Walden inversion).

83) If nucleophile center is same than nucleophilicity α basic strength.
$CH_3O^{\ominus} > H-O^{\ominus}$ [based on solvent]

84) By the removal of $-Cl_{(C)}$, due to formation of aromatic stable carbocation.

85)

Conceptual

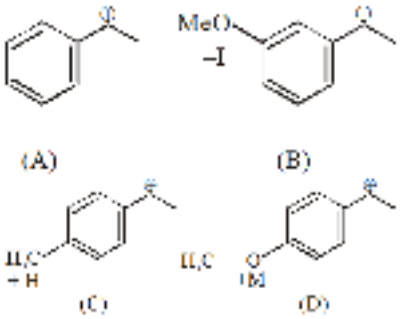
86) Given alkyl halide is primary so it shows S_N2 reaction instead of S_N1 reaction.
$CH_3O^{\ominus}$ is better nucleophile than $CH_3S^{\ominus}$ in polar aprotic solvent (DMF).

87) Hydrolysis will proceed by S_N1 mechanism.
Rate of S_N1 reaction depends on the stability of carbocation formed.
Comparing (I) and (II)
Benzyl cation is generally stable but in (II) the carbocation is also stabilised by +I group of CH_2CH_3.

Comparing (III) and (IV)
(III) will be less reactive-anti aromatic.
(IV) Will be more reactive as its carbocation is the most stable.

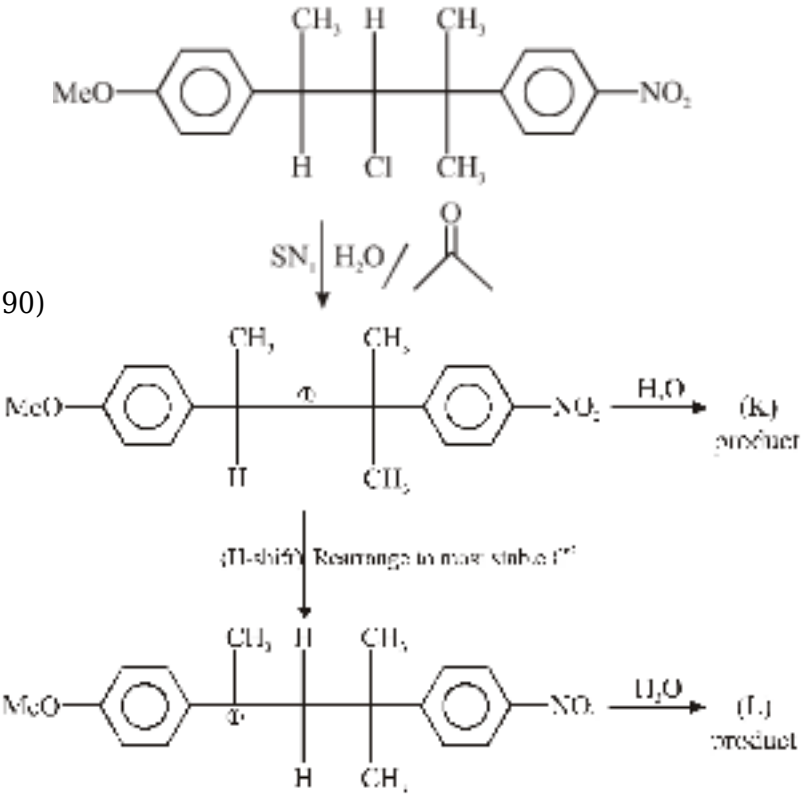
88) Leaving group ability a stability of conjugated acid

89) Stability of carbocation formed during SN^1



D > C > A > B stability of carbocation & reactivity of SN^1

90)



BIOLOGY

91) NCERT- (E) Page- 152

92) NCERT- (E) Page- 154

93) Ncert Pg No: 147

94) NCERT Pg. # 148-149

95) NCERT, # Human Health & Disease, Page 141, 4th para

96) NCERT PAGE NO.151

97) NCERT PG 147 and 149

98) Cancer, Allergy, and Rheumatoid arthritis are non-infectious diseases. They are not caused by pathogens but result from genetic, environmental, or autoimmune factors. The other groups (Options 2, 3, and 4) include diseases caused by pathogens such as bacteria, viruses, or fungi, and are therefore infectious.

The correct answer is : Option 1 (Cancer, Allergy, Rheumatoid arthritis).

99) IgG, IgM
NCERT (XII) Pg. # 151, Para–8.2.2

100) NCERT (XII) Pg. # 152, Para-8.2.4

101)

NCERT (XIIth) Pg. # 159

102)

NCERT Pg. # 131

103)

NCERT (XIIth) Pg. # 155 (para-8.3)

104) NCERT (XII) Pg. # 153,154

105)

NCERT Pg.No. 154

106)

The correct answer is Option 3.
Let's break down each statement:

A. **(i)** Interferons can prevent neighbor cells from viral infection: Correct. Interferons are

signaling molecules that alert neighboring cells about a viral infection, allowing them to initiate antiviral defenses.

B. **(ii)** lgM antibody always persists in monomeric form: Incorrect. lgM antibodies typically exist as pentamers (five units joined together) due to their structure.
C. **(iii)** Malaria is a non-infectious disease: Incorrect. Malaria is an infectious disease caused by parasites transmitted by mosquitoes.
D. **(iv)** Cancer is non-communicable: Correct. Most cancers are not contagious.

107) **Explanation:**

A. **A:** This step represents **reverse transcription.** The viral RNA is used as a template to synthesize viral DNA. This is a unique feature of retroviruses.
B. **B:** This step is the **integration** of the newly synthesized viral DNA (now called a provirus) into the host cell's nuclear DNA. The enzyme integrase is crucial for this process.
C. **C:** This step involves the **transcription and translation** of the viral DNA to produce new viral RNA and proteins, leading to the assembly of new virus particles.

Correct Answer : 1. B-lncorporation of viral DNA into host genome with integrase enzyme

108)

A. **Myasthenia gravis:**
This condition occurs when the body's immune system attacks the acetylcholine receptors at the neuromuscular junction, leading to muscle weakness.
B. **Rheumatoid arthritis:**
This autoimmune disease causes inflammation and joint pain due to the body attacking its own synovial tissue.
C. **Systemic Lupus Erythematosus (SLE):**
This systemic autoimmune disease can affect various organs and tissues throughout the body, including the skin, kidneys, and joints.

Why other options are incorrect:

A. **Gout:**
While gout is a painful inflammatory joint condition, it is not considered an autoimmune disease. It is caused by a buildup of uric acid in the joints.
B. **Muscular dystrophy:**
This group of genetic disorders affects muscle weakness and degeneration, but it is not caused by the immune system attacking the body's own tissues

109) NCERT Pg. # 135

110) NCERT (XIIth) Pg. # 146, 147

111) NCERT (XIIth) Pg. # 157, 158

112) NCERT (XII), Pg # 151

113) NCERT (XII), Pg # 151, fig.8.4

114) NCERT (XII) Pg. # 153, Para–8.2.5

115) NCERT (XII) Pg. # 153

116)

Explanation :
Here's why the other options are incorrect:

A. Tetanus, Tuberculosis, Measles: Measles is a viral disease, not bacterial.
B. Cholera, Typhoid, Mumps: Mumps is a viral disease.
C. Malaria, Mumps, Poliomyelitis: Malaria is caused by a protozoan parasite, mumps and poliomyelitis are viral diseases.

The correct answer is 2. Diptheria, Leprosy, Plague. All three of these diseases are caused by bacteria.

117) **Explanation:**
Autoimmune disorders occur when the immune system mistakenly attacks the body's own cells. These diseases differ from immunodeficiency disorders (like AIDS and SCIO), which weaken the immune system rather than causing self-attack.

I.D.D.M, also known as Type 1 Diabetes, is an autoimmune disorder where the immune system attacks insulin-producing beta cells in the pancreas. This leads to insulin deficiency and high blood sugar levels. Other options, like AIDS (caused by HIV) and SCIO (a genetic immunodeficiency), are not autoimmune diseases. Bronchial asthma is an allergic condition rather than an autoimmune disorder.

Correct Answer: 4. I.D.D.M (Insulin-Dependent Diabetes Mellitus)

118) **Explanation:**
The autoimmune disorders in the list are:

A. **Myasthenia gravis:** The body's immune system attacks acetylcholine receptors at neuromuscular junctions, causing muscle weakness.
B. **I.D.D.M. (Insulin-Dependent Diabetes Mellitus) or Type 1 Diabetes:** The immune system attacks the beta cells in the pancreas that produce insulin.
C. **Hashimoto disease:** The immune system attacks the thyroid gland, leading to hypothyroidism.

Correct Answer: 2. Three

119) NCERT XII, Page # 151

120)

This question focuses on the body's natural defense mechanisms that prevent pathogens from entering and establishing an infection.
Analysis of Each points that acts as physiological barrier
(A) Epithelium of Respiratory Tract: Incorrect.
(B) Monocyte: Incorrect.
(C) Tear: Correct.
(D) Skin: Incorrect.
(E) Saliva: Correct.
(F) Acid in Stomach: Correct.
(G) NK Cell (Natural Killer Cell): Incorrect.
Option 3 correctly includes C, E, and F (Tear, Saliva, and Acid in Stomach).

121) **Explanation:**
(i) Myasthenia Gravis: This disorder involves autoantibodies targeting **acetylcholine receptors** at neuromuscular junctions, causing muscle weakness.
(ii) Pernicious Anemia: This condition results from autoantibodies against **intrinsic factor**, a protein needed for vitamin B12 absorption.
(iii) IDDM (Insulin-Dependent Diabetes Mellitus) / Type 1 Diabetes: This is an autoimmune attack on the **beta cells of the pancreas**, which produce insulin.
(iv) Multiple Sclerosis: This disease involves autoantibodies attacking the **myelin sheath** that surrounds and protects nerve fibers in the central nervous system.

Correct Answer : 1. (i)-d, (ii)-c, (iii)-b, (iv)-a

122) NCERT Pg No. # 151 (Para 8.2.2)

123) **Explanation:**

A. A (lgE): lgE antibodies are central to allergic reactions. They bind to mast cells.
B. **B (Histamine):** Histamine is a chemical released by mast cells during an allergic reaction. It causes many allergy symptoms.
C. **C (Running Nose):** A runny nose is a common symptom of allergies, caused by histamine release.
D. **D (Steroids):** Steroids (corticosteroids) are often used to treat allergies because they reduce inflammation.

Correct Answer: 4. lgE, Histamine, Running Nose, Steroids

124) NCERT XII, Page # 148, Fig:8.1

125)

The diseases caused by bacteria in the list are:
1. Pneumonia (caused by *Streptococcus pneumoniae* or *Haemophilus* influenzae)
2. Typhoid (caused by *Salmonella typhi*)
3. Tetanus (caused by *Clostridium tetani*)

126) **Explanation :**

A. Symptoms : Slowly developing chronic inflammation of the organs.
B. Caused by Wuchereria malayi.
C. This is correct; Wuchereria malayi causes filariasis, and it leads chronic inflammation.

Thus, the correct option is: 3.

127) NCERT-XII, Pg. # 135, 136

128)

Here's a breakdown of the correct answer:
Antibody Structure:
• Variable Region: This region of the antibody is responsible for recognizing and binding to specific antigens. Each variable region typically contains one disulfide bond.
• Constant Region: This region determines the class of the antibody (e.g., lgG, lgA, lgM) and interacts with other components of the immune system. Each constant region typically contains three disulfide bonds.
Therefore, the number of disulfide bonds present on one variable region of the light chain and one constant region of the heavy chain, respectively, is 1, 3.
So, the correct answer is Option 4.

129)

Syphilis, Gonorrhoea, genital Herpes, chlamydiasis, genital warts, Trichomoniasis, hepatitis-B, AIDS are STD

130) NCERT Pg# 149 (E)

131) **Explanation :**
(2) A = Interferons, B = Clinical infection, C = vaccine, D = Colostrum, E = Antiserum: This option aligns perfectly with our analysis:

A. A (Innate Immunity): Interferons
B. B (Natural Active Acquired Immunity): Clinical infection
C. C (Artificial Active Acquired Immunity): vaccine
D. D (Natural Passive Acquired Immunity): Colostrum
E. E (Artificial Passive Acquired Immunity): Antiserum

The correct option is 2.

132) Statement I: Correct. Cellular oncogenes (c-onc) or proto-oncogenes are present in normal cells and can lead to cancerous transformations when activated under certain conditions. Statement II: Incorrect. Techniques like biopsy and histopathological studies are indeed very useful for detecting cancers, including those of the internal organs. Additional techniques such as radiography, CT, and MRI also play a significant role in cancer detection.

133) The assertion is true as pneumonia can cause the lips and fingernails to turn gray to

bluish due to poor oxygenation of blood. The reason is also true and is the correct explanation for the assertion because when the alveoli are filled with fluid, it prevents efficient gas exchange, leading to reduced oxygen in the blood and hence the discoloration observed in severe cases of pneumonia.

134) Statement I: Correct. The fungi belonging to the genera Microsporum, Trichophyton, and Epidermophyton are indeed responsible for ringworms, which are a common infectious disease in humans. Statement II: Incorrect. Ringworms can be acquired from various sources, including soil, as well as by using towels, clothes, or the comb of infected individuals. Therefore, it is not true that ringworms are acquired only through personal items of infected individuals.

135) Statement I: Correct. Vaccines introduce an antigenic protein of a pathogen or an inactivated/weakened pathogen into the body, leading to the production of antibodies and the generation of memory B and T-cells that help in recognising and fighting the actual pathogen upon future exposure. Statement II: Correct. In cases where an immediate immune response is necessary, such as with tetanus, it is more effective to directly inject preformed antibodies or antitoxin, which contains antibodies to the toxin, rather than waiting for the body to mount an immune response to a vaccine.

136) NCERT-XII, Pg. # 89

137) NCERT XII Pg # 90, 91

138) NCERT (XII) Pg. # 88

139) NCERT-XII, Pg. # 88

140) NCERT-XII, Pg. 83

141) NCERT-XII, Pg. # 75

142)

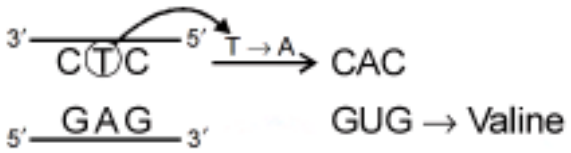
NCERT XII Pg. # 89
If normal is aa then affected will be Aa.

143) Hint: X linked disorders show cris-cross inheritance.
Sol.: Given pedigree is applicable for X linked disorders like colourblindness and haemophilia. Genotype of female in first generation can be XCX or X^hX. Pedigree analysis is not done for disorder caused by genomatic mutation.

144) In autosomal recessive disorders both unaffected parents can have affected child so, both parents are heterozygous.

145)

NCERT-XII, Pg. # 72

146) Substitution of Glutamine 

147)

NCERT XII Pg # 88

148)

NCERT-XII, Pg. # 89

149)

NCERT-XII, Pg. # 89, Fig. 5.14

150)

NCERT XII, Pg # 89; Para : fig. 5.14, b

151) NCERT pg.no. 113

152) Allen module question 141, pg.no. 148

153) Allen module question 150, pg.no. 149

154) Allen module question 161, pg.no. 150

155) NCERT Pg # 90,91

156) NCETT XII Pg.No. 113 & Module Pg. No. 127

157) NCERT Pg. # 86

158) NCERT Pg. # 87

159) NCERT-XII, Pg # 114

160)

NCERT XII Pg. Eng-113
NCERT XII Pg. Hindi-124

161)

NCERT Page No. 76

162) NCERT Page No. # 113

163) Ncert Page 74 and 97.

164) NCERT Pg. # 106

165) Ncert Pg No: 90_ 2021-22

166) NCERT Pg. # 145

167) Module-1 Pg. # 130

168) NCERT (XI) Pg. # 00

169) NCERT Pg. # 00

170) NCERT XI Pg # 145

171) NCERT XII Pg # 106

172) NCERT XI Pg.# 145 fig. 9.1

173) NCERT XI Pg. # 145

174) NCERT (XI) Pg. # 145, Fig. 9.1

175) NCERT-XI Pg. # 145

176) NCERT (XIth) Pg. # 97, Para # 3

177) NCERT Pg. # 152 1st para

178) Module-1 Pg # 132

179)

NCERT-XII, Pg # 96

180) NCERT (XI) Pg. # 144, IInd Para.